
FRC properties at fresh and hardened states

How Fiber Reinforced Concrete can be integrated as an economical, sustainable, and durable aspect of our infrastructure

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**Barzin Mobasher
Arizona State University**

FRC Components

- Fiber reinforced concrete (FRC) has the same ingredients as plain concrete, with the addition of short fibers.
- Portland cement, aggregates, water, admixtures, and fibers. The fibers are distributed throughout the mixture as the mixing proceeds and provide a three dimensional reinforcement throughout.
- A variety of fiber types and geometries are available for use in concrete and they can be added at various dosages.
- The amount of fiber addition and the type of fiber depend on the specific application.



Variables Controlling the Behavior of FRC at early and mature age

- Ingredients
- Environment, age
- Mechanisms
- Properties,
- Changes due to environment and properties evolution

Factors affecting the properties

- Initial mix proportions and design of components
- Early age property development, rheology, consolidation, pumping, maturity
- Age related properties, early age to maturity and durability
- Mechanisms, bond, interface, tensile, compressive
- Environment, Durability, corrosion, expansion and shrinkage
- Loading, cracking, propagation, Fatigue, aging, creep



Properties of Fresh and Hardened FRC

Start from ordinary concrete technology, 1960-1970



High performance concrete, 1990, durability



Ultra-High-performance concrete, 120 MPa, significant ductility



Use Mechanics and Materials Science, Add in the fibers, characterize and incorporate the tensile response in the design.



Applications of FRC in Serviceability based Design Applications

- Phenomenal growth of the market for fiber reinforced concrete as a key sustainability measure.
 - Economy, labor, materials characteristics, and performance
 - Type, length, and content of fibers, affects the tensile response of FRC.
 - Serviceability based design opportunities:
 - Ductility, durability, crack width, stiffness, cracked section modulus, Shear
 - A hybrid approach of combining reinforcement and fibers
 - Minimum reinforcement requirements.
 - Software based Design
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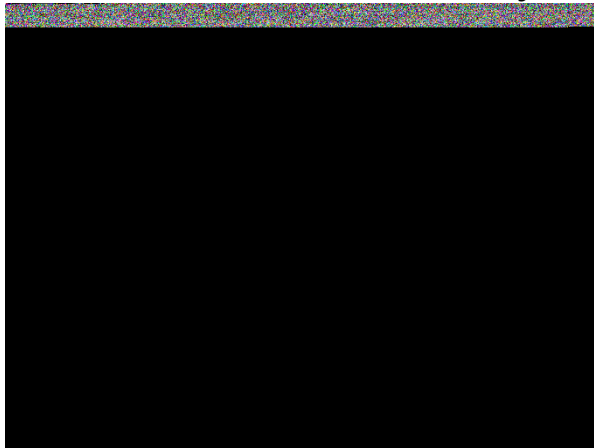
Whitney Stress Block



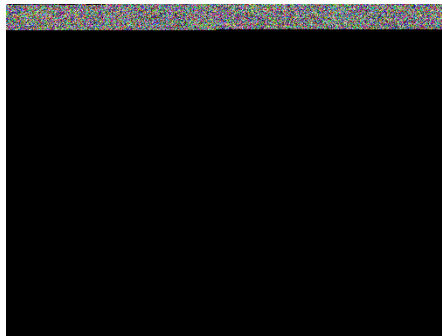
- Ignore Tensile contribution of concrete
- Unable to be extended to incorporate tensile stresses in the formulation
- Empirical assumptions representing a single point of limit state in the loading history making design optimization impractical
- Unable to incorporate other failure criteria such as crack width, or stiffness
- Impracticality of the concrete cover under the service load cracking
- Cracking under the service load
- Inability to characterize the effective stiffness under service conditions

Why Should we use Fibers and Textiles?

Strengthening at the microstructure level will have a profound impact at the macro-structural level strength and durability

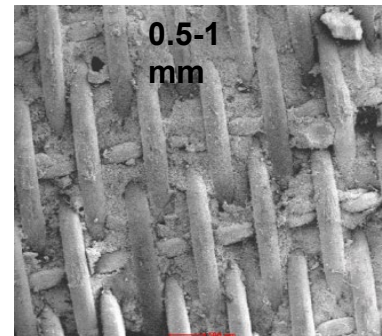


Reinforcing Bars:
Macro-structure
reinforcement



Fiber reinforced Concrete (FRC), reinforced at the meso-structure with $\frac{1}{2}$ " long fibers, $\frac{1}{2}$ " (10mm) millimeters apart

**Fiber reinforced
Concrete:** Meso-structure
reinforcement



Textile reinforced Concrete (TRC), reinforced at the micro-structure with long fibers, woven, and only millimeters apart

**Textile Reinforced
Concrete:** Micro-structure
reinforcement

Materials Design, Mechanical Properties, Structural Design

Materials Design:

- Ingredients, admixtures, fibers, processing
- Applications that require a realistically measured serviceability criteria in terms of crack width, deflection, curvature, crack spacing, stiffness, residual strength, or fatigue.
- Develop standard structural design procedures by avoiding the high costs of proprietary mixes, finite element modeling, or full scale testing.

Mechanical Properties

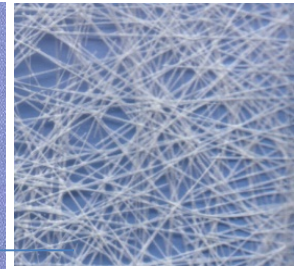
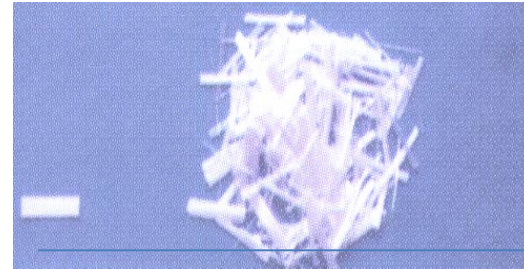
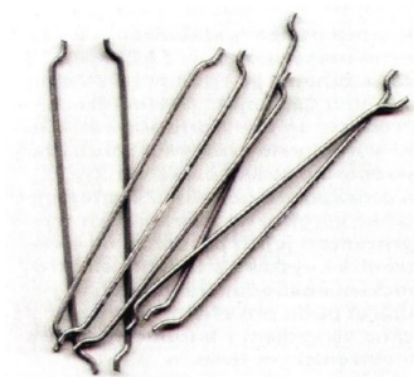
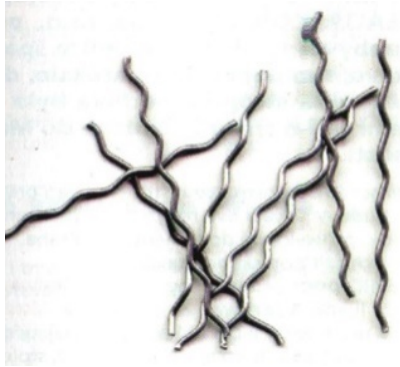
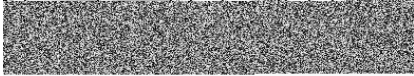
- Parametric linearized materials model
- Cross sectional analysis based on stress and strain diagrams
- Back calculation of tensile responses and extraction of tensile properties

Structural Design

- Serviceability state design: curvature, deflection, post-crack stiffness
- Design recommendations: moment capacity, minimum reinforcement ratio, serviceability limits



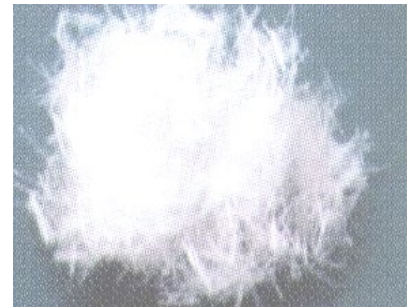
Fiber types



Glass fibers (Alkali Resistant)



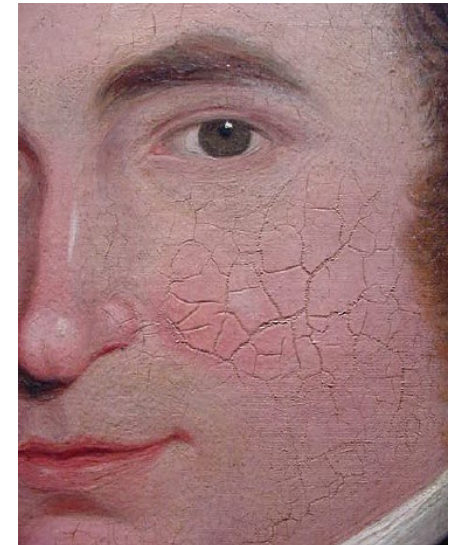
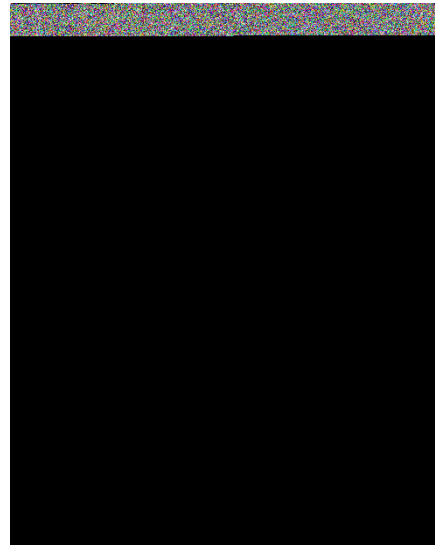
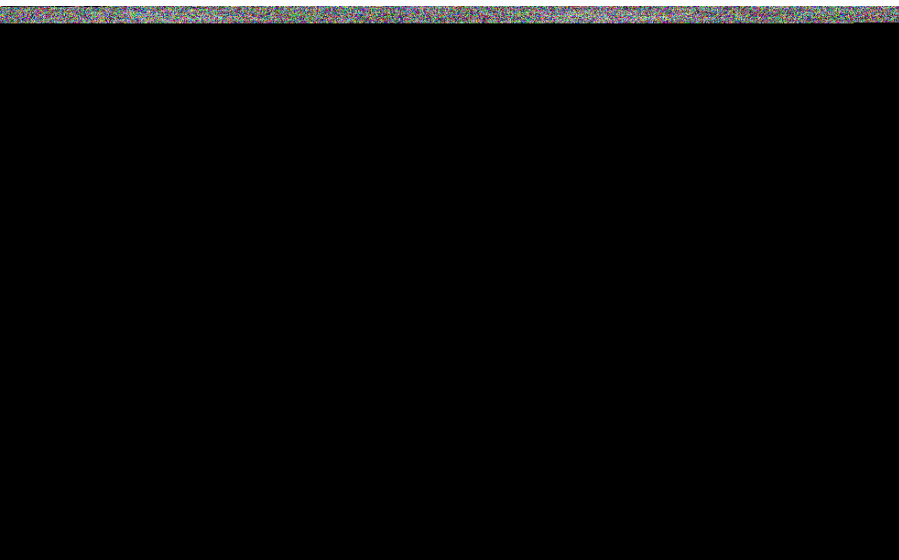
Steel fibers



Synthetic fibers (micro- and Macro)

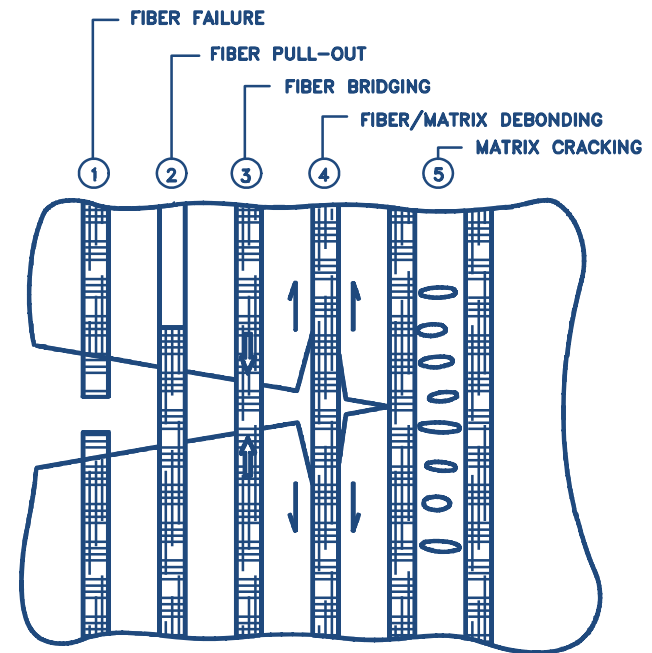
Cracking Phenomenon

- Brittle materials are weak under tensile (and flexural) loads. They are also very sensitive to sharp notches and internal flaws which may be the starter points for cracks.
- The main function of the fibers is to support concrete's integrity after cracking
- Crack formation is a dominant form of failure in many consumer products, structural materials, art, and even natural events are attributed to it.



Role of fibers in FRC

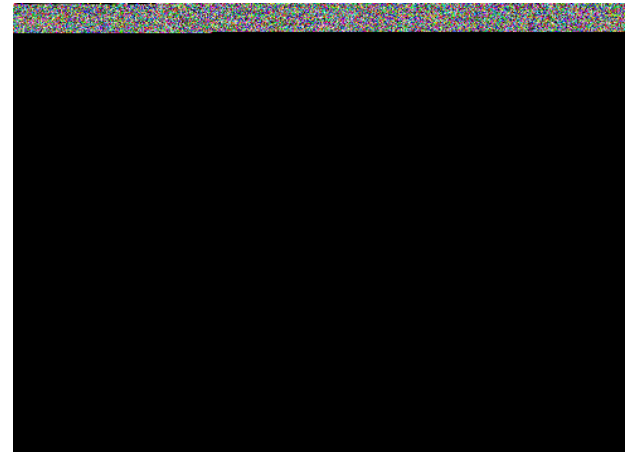
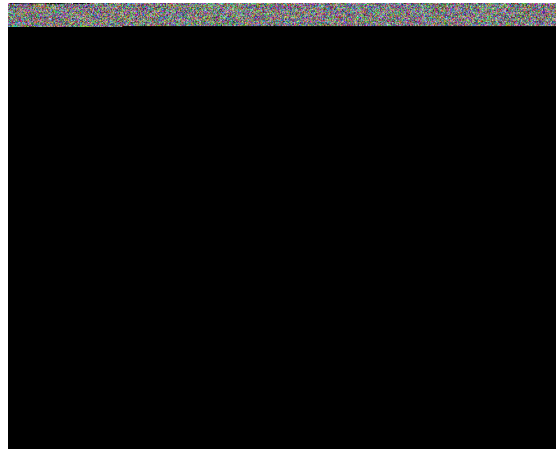
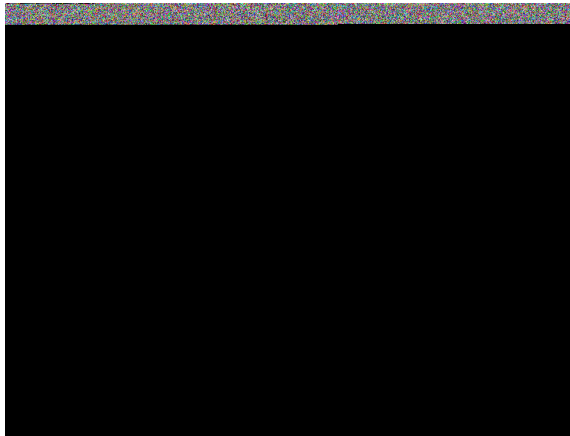
- When concrete is loaded in tension, short fibers stretch due to tensile forces and carry some stress. Once the concrete cracks, these fibers do not necessarily fail, so they end up bridging the tensile (or flexural) cracks, and start to carry more stresses. The closer the fibers are to a small crack, the sooner they can stop it from growing uncontrollably. That is why large fiber contents increases the chance of fiber bridging the cracks on their incipient formation. The stages involved in FRC failure is shown in the figure: 5) cracks form in cement matrix, 4) debonding and sliding between fiber and matrix, 3) bonded fiber bridging the cracks, 2) frictional sliding and fiber pullout, and 1) fiber failure under tension.



Ref: "Fracture Mechanics" by
T.L. Anderson, CRC Press,
1991.

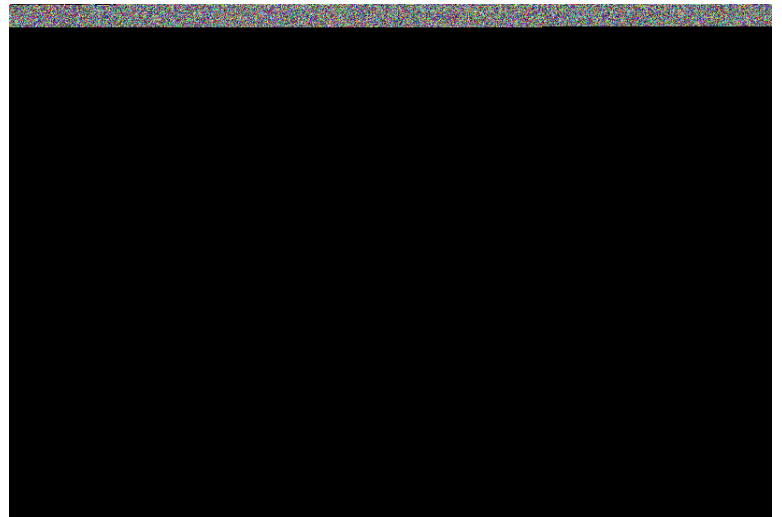
Microstructure of FRC

- Fiber-reinforced concrete has three distinct phases:
 - Matrix phase
 - Interface
 - Fiber phase
- Interface in FRC is a distinct layer between the phases that bonds with the hydrated cement paste and fiber. Interfacial transition zone (ITZ) has higher porosity, similar to the ITZ between aggregates and cement paste, and is rich in Calcium hydroxide. This zone is relatively weak and can fail and allow the fibers to slip (de-bonding).



Interface

- Bond of fiber is associated with three types of mechanisms:
 - Chemical Bond
 - Not very common, could result in fiber degradation
 - Mechanical Bond
 - Affected by fiber shape (higher in twisted, curved, fibrillated fibers)
 - Frictional Bond
 - Causes slip-pullout and energy dissipation
 - Is a weak sacrificial layer
 - Shrinkage is an example of frictional bond
- The bond should not be too strong as the failure of interface is beneficial to many degrees. Strong interfaces can make very brittle composites. Weaker interfaces make ductile composites.
- Major Mechanisms
 - Stress Transfer
 - Debonding, delamination and fiber pullout



Strong Solids, A. Kelly

Testing Fresh FRC

- The majority of tests that are performed on fresh concrete are applicable to fresh fiber-reinforced concrete.
- These include density, air content, setting time, and temperature changes after casting and are typically performed for quality control (QC) purposes.
- Conventional ASTM test methods can be used for determining these properties. The workability testing of fresh FRC and flexural properties of FRC are the main outliers to the testing.

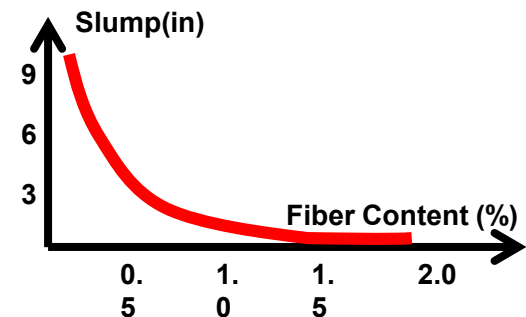
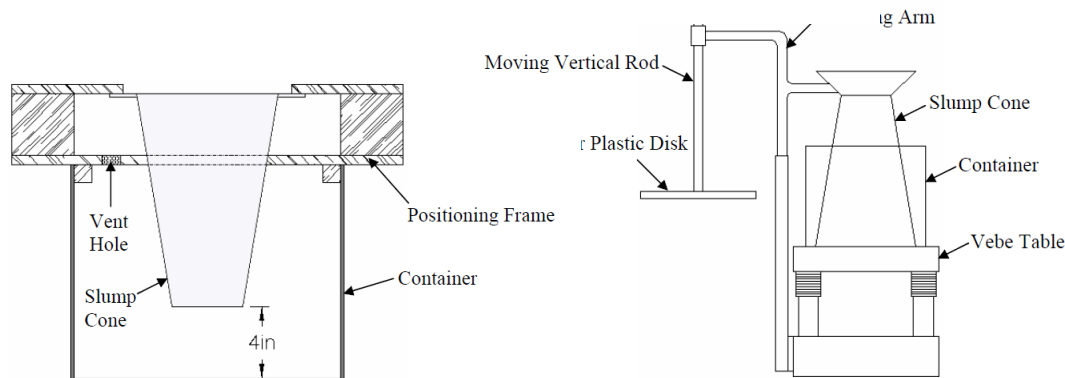
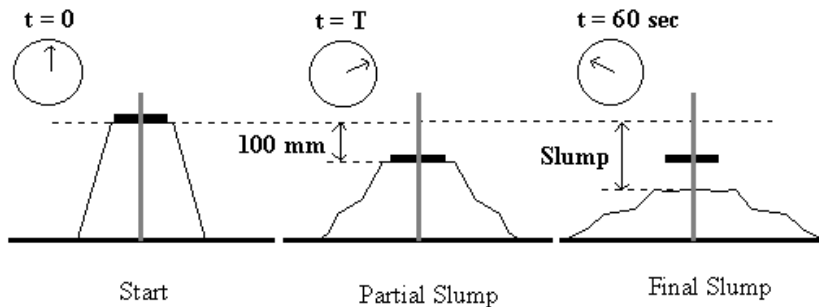


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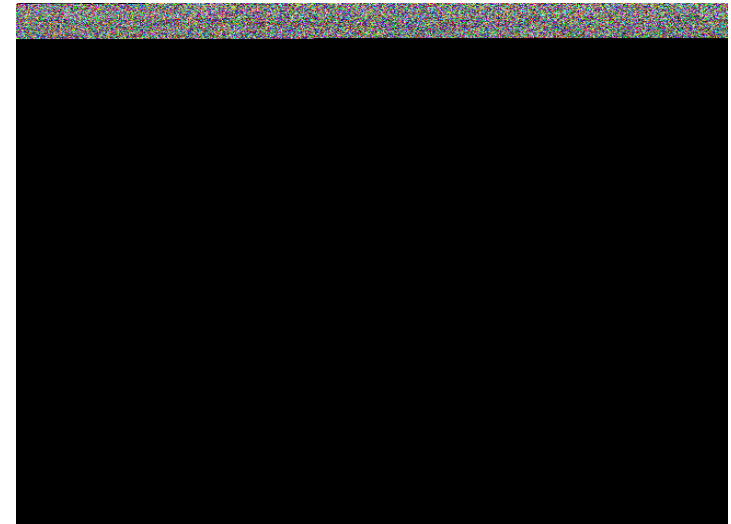
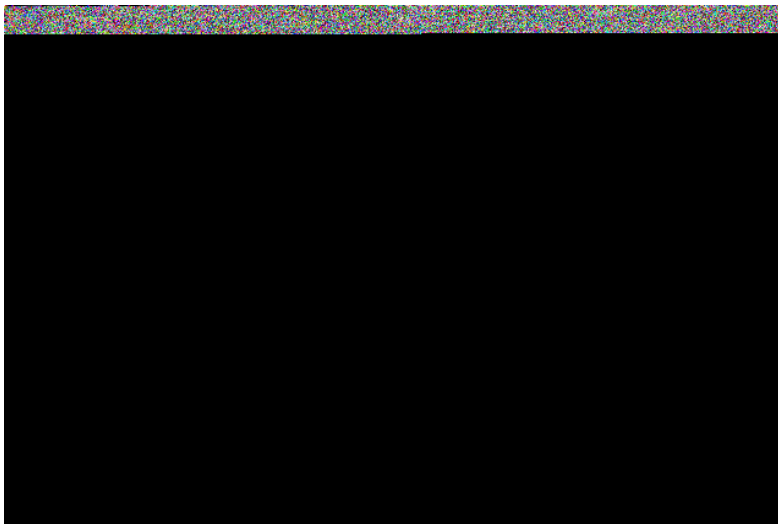
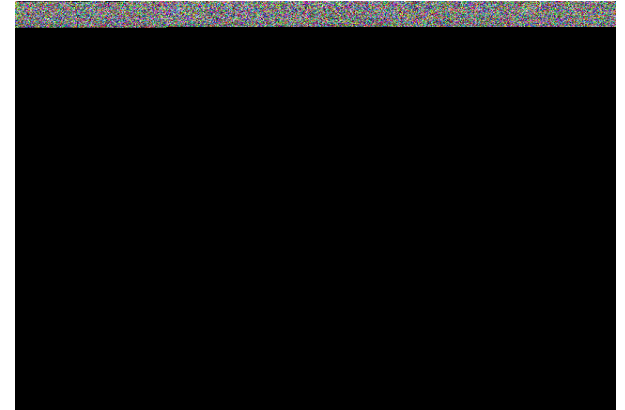
Slump Test

- The slump test ASTM C 143 used for measuring the slump of fresh FRC.
- May not be a good indicator of workability for FRC
- used as a quality control measure to monitor the batch-to-batch consistency.
- FRC without plasticizers may show very small (or zero) slump at higher fiber contents if not compacted with mechanical vibration.
- Inverted slump test
- Vebe time



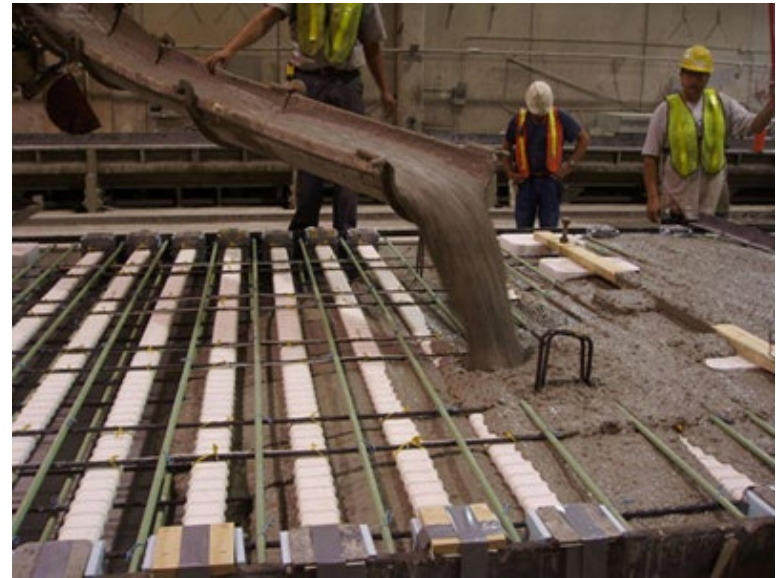
Rheology Testing of FRC

- Rheology: study of material flow or deformation under stress.
- Rheological parameters describe the ease with which the material responds to workability, flow, pumpability, placement, compaction, and finishing.
- FRC workability tests are similar to normal concrete, mainly the well-known slump test.
- FRC typically requires mechanical vibration (external) for proper consolidation. Plasticizers are used to improve the workability using available modern superplasticizers, rheology and viscosity modifiers.



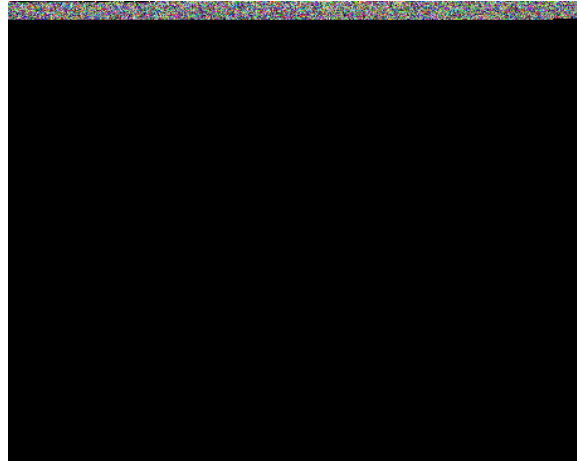
Self-Consolidating Concrete, SCC

- Superplasticizers are used to give FRC self-consolidating or self-flowing properties.
- Fibers will have a good alignment when used with SCC.
- Mechanical vibrations is eliminated for SCC.
- Applications where compacting fresh concrete is difficult due to the reinforcement level, element size, or fiber content.



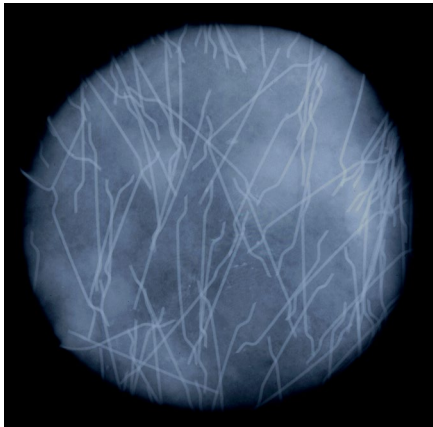
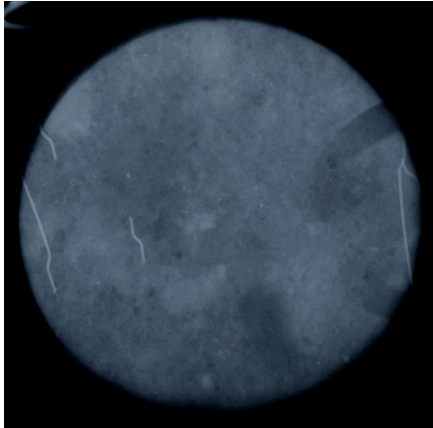
Self-Consolidating Concrete

- Special rheology tests are required for testing SCC.
- Self-flowing characteristic.
- ASTM C 1611-09 active standard test method for measuring the slump flow of SCC.
- ASTM C 1621-09 test method for measuring the passing ability of SCC using J-ring.
- ASTM C 1610-10 potential segregation of SCC using column technique.
- typically used for quality control of fresh SCC (with or without fiber reinforcement).

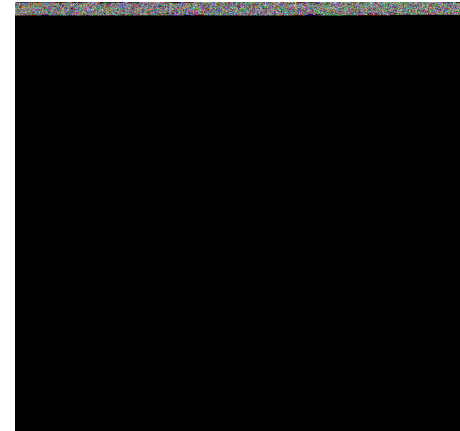


Fiber Distribution in Concrete

Vibrated FRC
(steel fibers tend to sink)



Self-Consolidating FRC
(more uniform distribution)

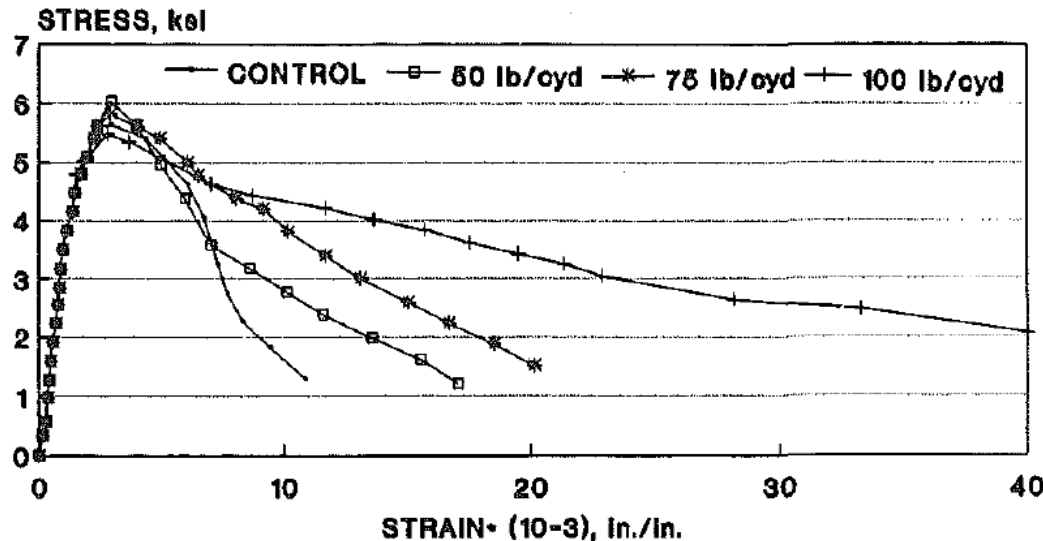
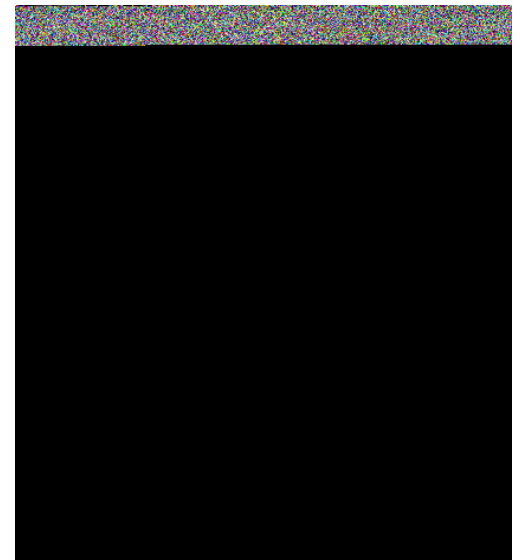
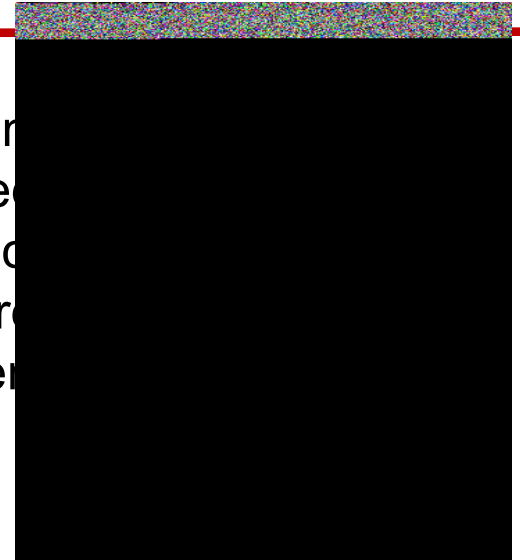


X-ray image
from the top of
the drilled
core

X-ray image
from the
bottom of the
drilled core

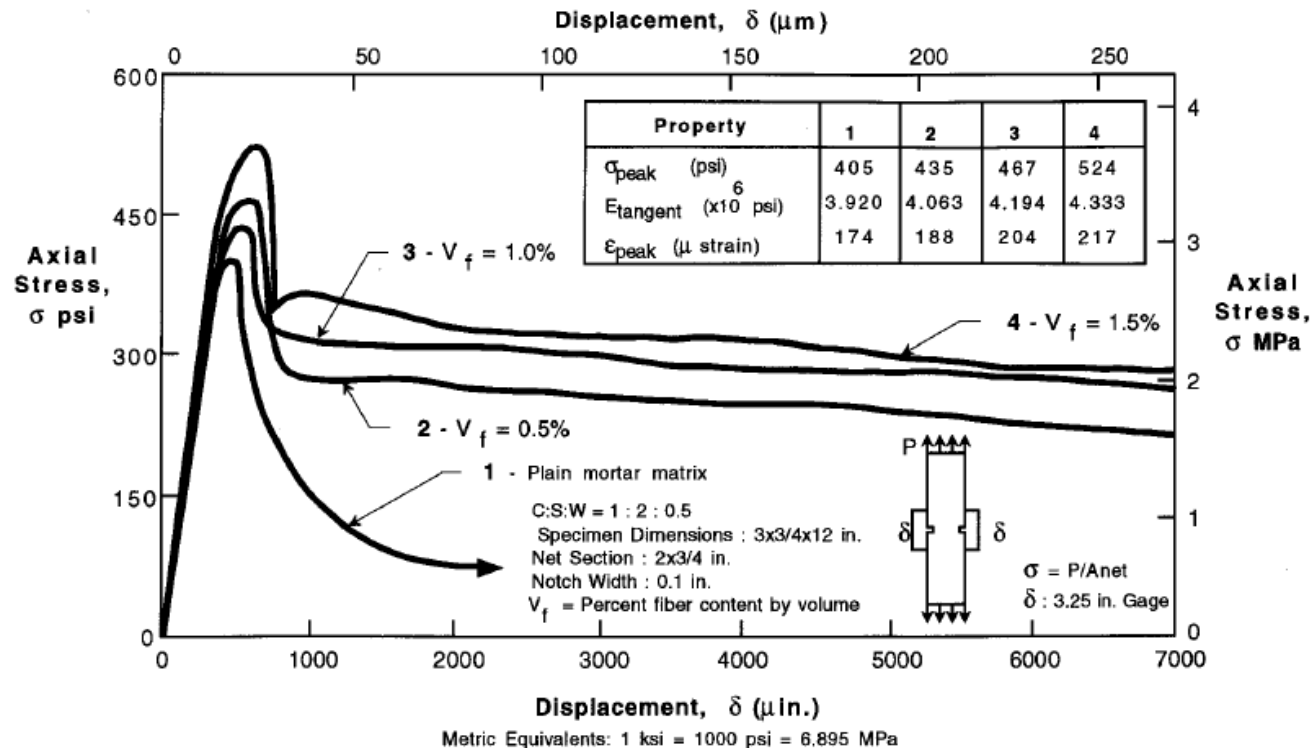
FRC Material Properties: Compression

- Fibers can slightly increase the compressive strength. The effect is on the mode of failure. Plain (unreinforced) concrete fails under brittle (sudden) failure. The integrity of concrete is maintained if short fibers are added and the failure is more ductile. Fibers provide a small level of internal confinement, which delays failure as the internal cracks are held together.



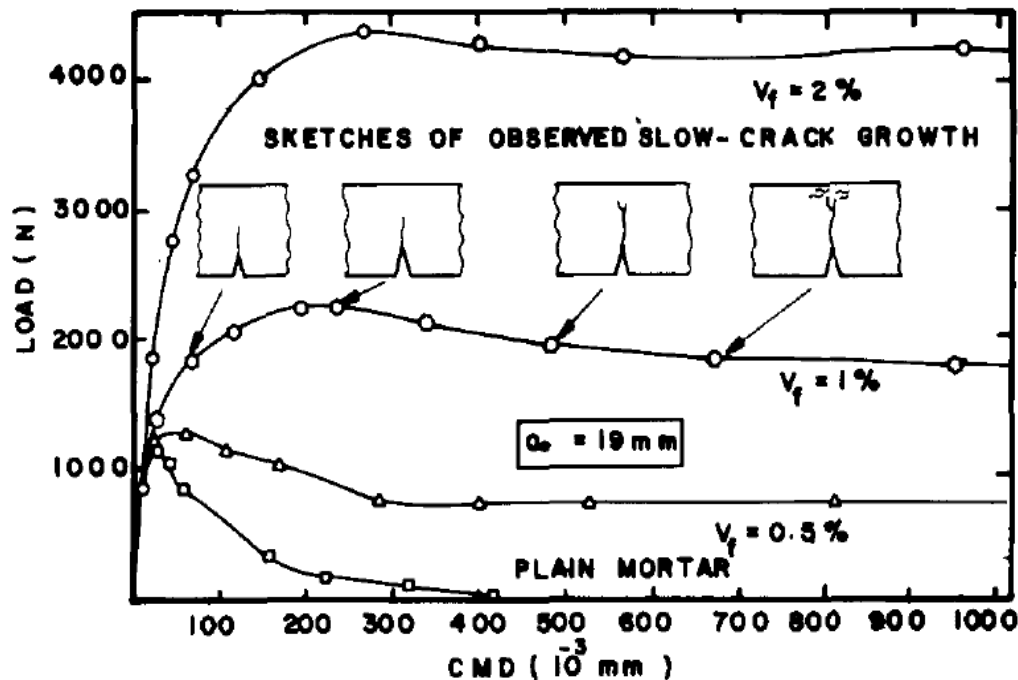
FRC Material Properties: Tension

- Fibers can increase both the tensile strength and tensile toughness when used at sufficiently high volume fractions. The main effect of fibers in the tensile response is in increasing the toughness as they change the mode of failure from a brittle fracture to ductile behavior. The figure shows tensile response of FRC with various amounts of steel fibers



FRC Material Properties: Bending

- Fibers can increase the flexural strength, but their main effect is on the mode of failure: brittle to ductile. Therefore their contribution is apparent after the concrete cracks. Depending on the type, length, and content of fibers, the level of ductility changes. The figure shows the flexural response of FRC with various amounts of steel fibers in comparison with mortar (plain concrete). Note that as the fiber content increases the post crack response approaches a response similar to plastic be



Fiber Pull-out Mechanism

- Fiber pull-out is one of the predominant mechanisms. This figure shows the the fiber resists the pullout force much piece of wood. Once the peak load is reached and dissipates more energy as it slides depends on various parameters such as anchorage, ends, roughness, friction



Composite Materials Approach

- Elastic Response
- Rule of mixtures, volume averaging, mean field, self consistent methods
- Average properties
- distinct phases of the components and properties



Voigt Model (Parallel or Iso-strain)

- Equilibrium $\sigma A = \sigma_1 A_1 + \sigma_2 A_2$
- Compatibility $\varepsilon = \varepsilon_1 = \varepsilon_2$
- Constitutive relationship $\sigma = E\varepsilon$

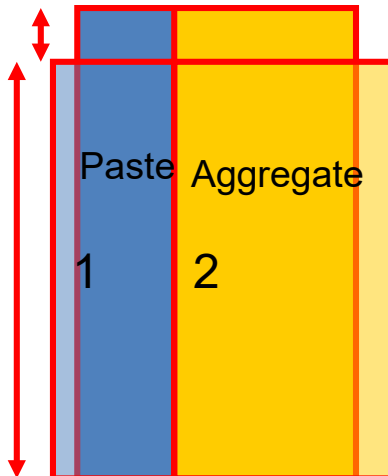
$$E\varepsilon A = E_1\varepsilon_1 A_1 + E_2\varepsilon_2 A_2$$

replace V and A

$$EV = E_1V_1 + E_2V_2$$

or

$$E = E_1V_1 + E_2V_2$$



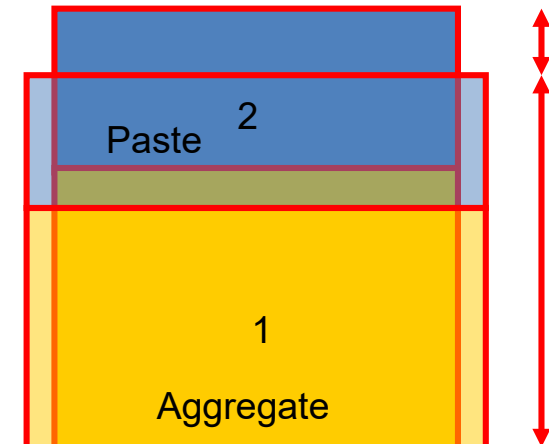
Kelvin Model, (Series or Iso-stress)

- Equilibrium $\sigma = \sigma_1 = \sigma_2$
- Compatibility $\varepsilon = \varepsilon_1 + \varepsilon_2$
- Constitutive relationship $\sigma = E\varepsilon$
- Total Elongation $\varepsilon L = \varepsilon_1 L_1 + \varepsilon_2 L_2$

- Average Stress computed from:

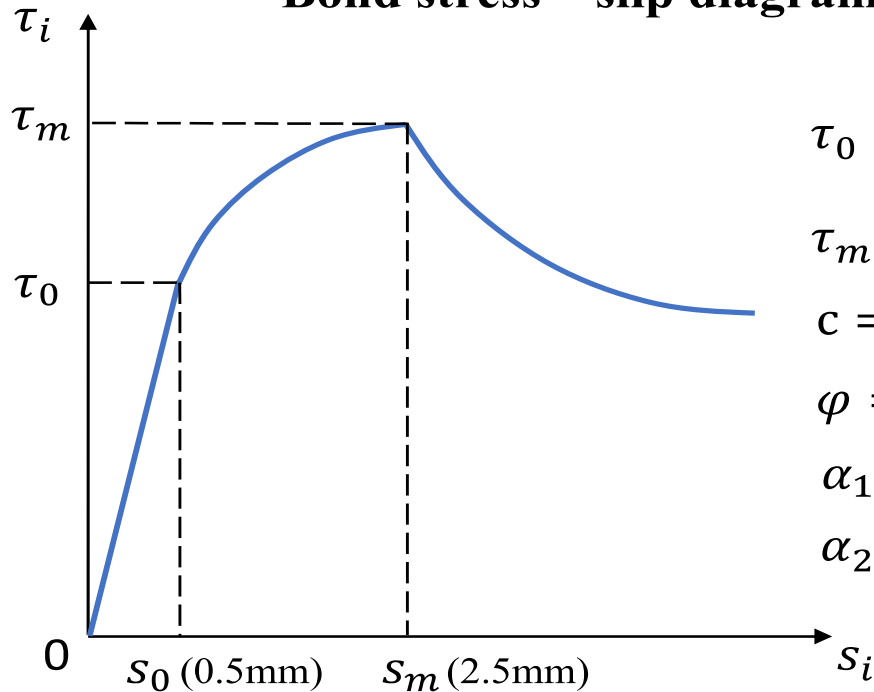
$$C_i = \frac{L_i}{L} = \frac{V_i}{V} \quad \frac{\sigma}{E} = \frac{L_1\sigma_1}{E_1} + \frac{L_2\sigma_2}{E_2}$$

$$\frac{1}{E} = \frac{C_1}{E_1} + \frac{C_2}{E_2}$$



Interface problems, rebars, fibers, multiple layers

Bond stress – slip diagram



$$\tau_0 = \tau_m \left[\frac{s_0}{s_m} \right]^{\alpha_1}$$

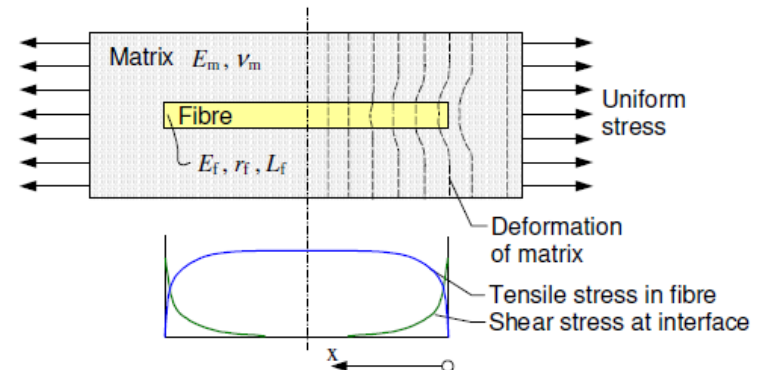
$$\tau_m = \sigma_n \tan \varphi + c$$

$$c = 1 \text{ MPa}$$

$$\varphi = 37^\circ$$

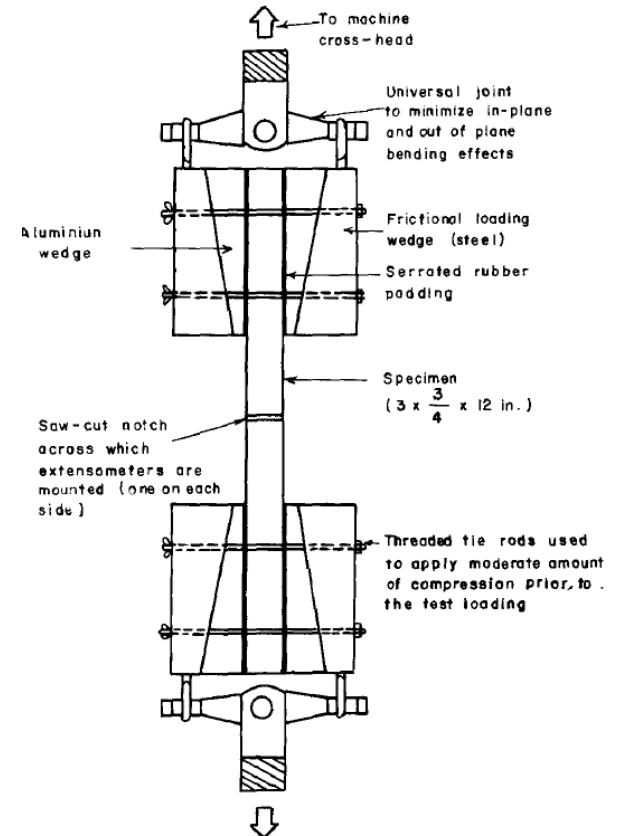
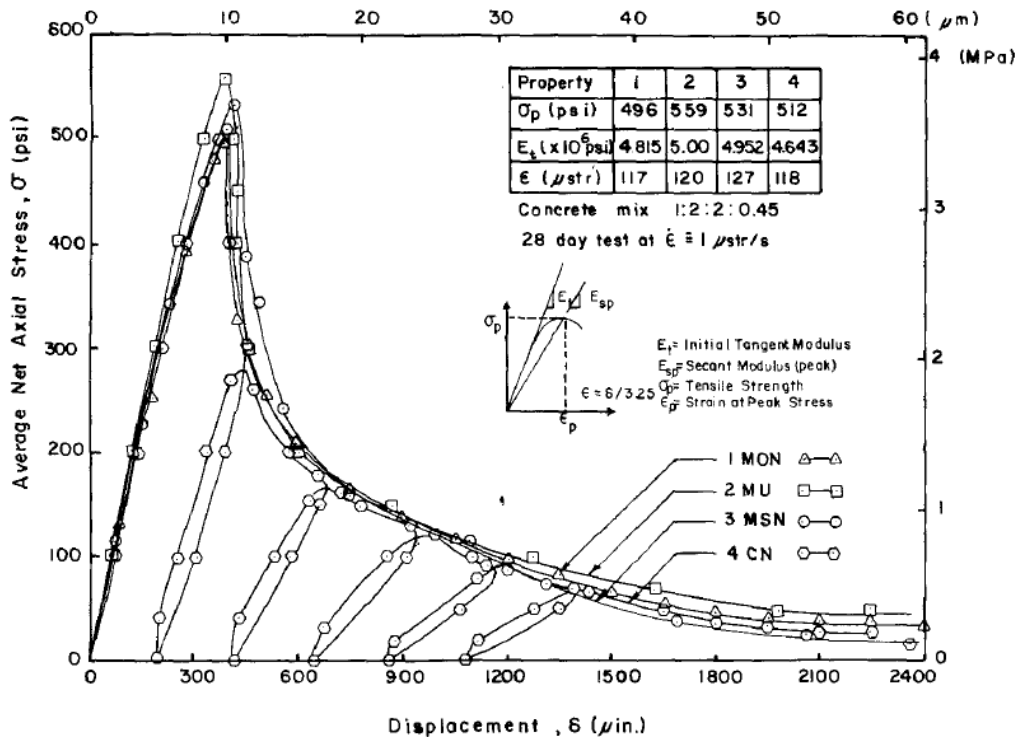
$$\alpha_1 = 1$$

$$\alpha_2 = 1$$



Softening in Tension

- Gopalaratnam and Shah, 1985, ACI Journal



Gopalaratnam and Shah

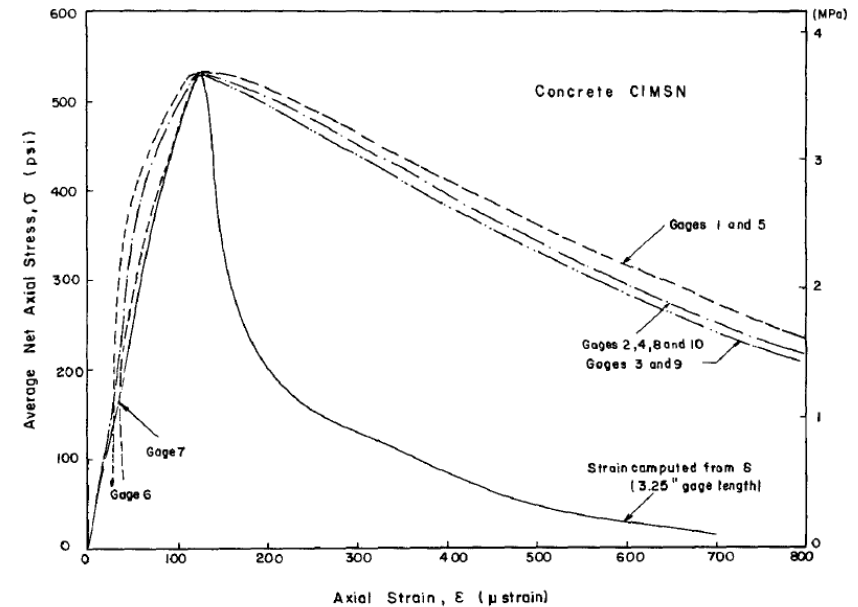
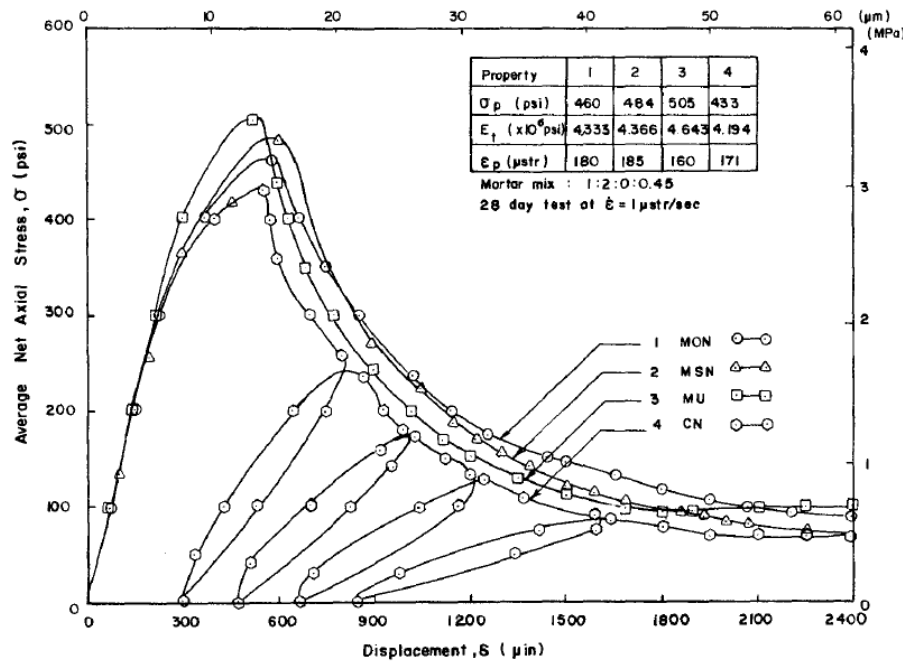


Table 2—Average material properties of the various mixes tested

Mix details				Tension tests						Compression tests			
Materials	Series	Mix proportions (by weight) cement:sand:aggregate:water	Age, days	Strength σ_p , psi	Initial modulus E_t , $\times 10^6$ psi	Strain at σ_p , ϵ_p , μ strain	$\frac{E_t \epsilon_p}{\sigma_p}$	Fracture toughness G_{fc} , lb/in.	Fracture energy G_f , lb/in.	Strength f'_c , psi	Initial modulus E_c , $\times 10^6$ psi	Strain at f'_c , ϵ_c , μ strain	$\frac{E_c \epsilon_c}{f'_c}$
Concrete	C1	1:2:2:0.45	28	525	4.853	120	1.109	0.050	0.322	6364	4.410	1843	1.277
	C2	1:2:2:0.60	28	439	4.729	126	1.357	0.037	0.297	4836	4.240	1800	1.578
	C3	1:2:2:0.60	3	326	3.966	97	1.180	0.024	0.216	—	—	—	—
	C4	1:2:2:0.45*	28	549	5.375	132	1.292	0.050	0.341	Same as in Row 1			
Mortar	M1	1:2:0:0.40	28	540	4.663	163	1.408	0.055	0.537	6790	4.547	2352	1.575
	M2	1:2:0:0.45	28	471	4.384	174	1.620	0.044	0.476	5600	4.244	2420	1.833
	M3	1:2:0:0.50	28	405	3.920	174	1.684	0.038	0.415	—	—	—	—
Paste	P1	1:0.5:0:0.35	28	471	3.490	198	1.467	0.057	0.404	—	—	—	—

*Specimen thickness = 1.5 in. All others 0.75 in. thick. 145 psi = 1 MPa, 1 lb/in. = 175.2 N/M.

Softening response

Ascending portion — A simple expression that has been used for defining compressive behavior of confined and unconfined concrete¹² is recommended for the uniaxial tensile stress-strain curve up to the peak stress value

$$\sigma = \sigma_p \left[1 - \left(1 - \frac{\epsilon}{\epsilon_p} \right)^A \right] \quad (3)$$

where

- σ = tensile stress
- σ_p = peak value of σ
- ϵ = tensile strain
- ϵ_p = value of ϵ at σ_p

$$A = E_t \epsilon_p / \sigma_p$$

E_t = initial tangent modulus

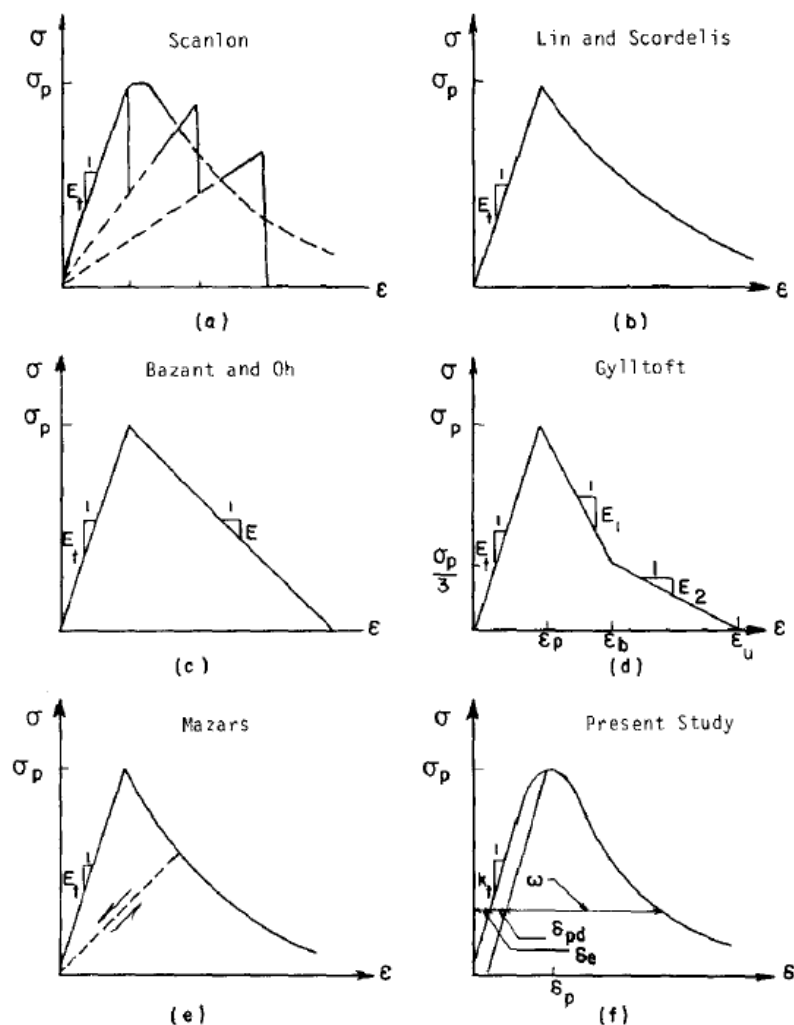
Note that A is always greater than one for any material exhibiting pre-peak nonlinearity (Table 2). As a result, the slope of the stress-strain curve [Eq. (3)] is always zero at the peak.

Descending portion — The relationship between post-peak stress and crack width is given by

$$\sigma = \sigma_p (e^{-k\omega\lambda}) \quad (4)$$

where ω is crack width and k and λ are constants. For the sake of continuity at the peak, it is desirable that λ be greater than one. Thus, although a value of λ less than one gave a better fit to the experimental data, $\lambda = 1.01$ was assumed here. For this value of λ , a value of $k = 1.544 \times 10^{-3}$ (when ω was expressed in $\mu\text{in.}$) gave





$$\delta = \delta_e + \delta_{pd} + \omega$$

$$= \frac{\sigma}{K_t} + \left(\delta_p - \frac{\sigma_p}{K_t} \right) + \left(-\frac{1}{k} \ln \frac{\sigma}{\sigma_p} \right)^{1/\lambda} \quad (5)$$

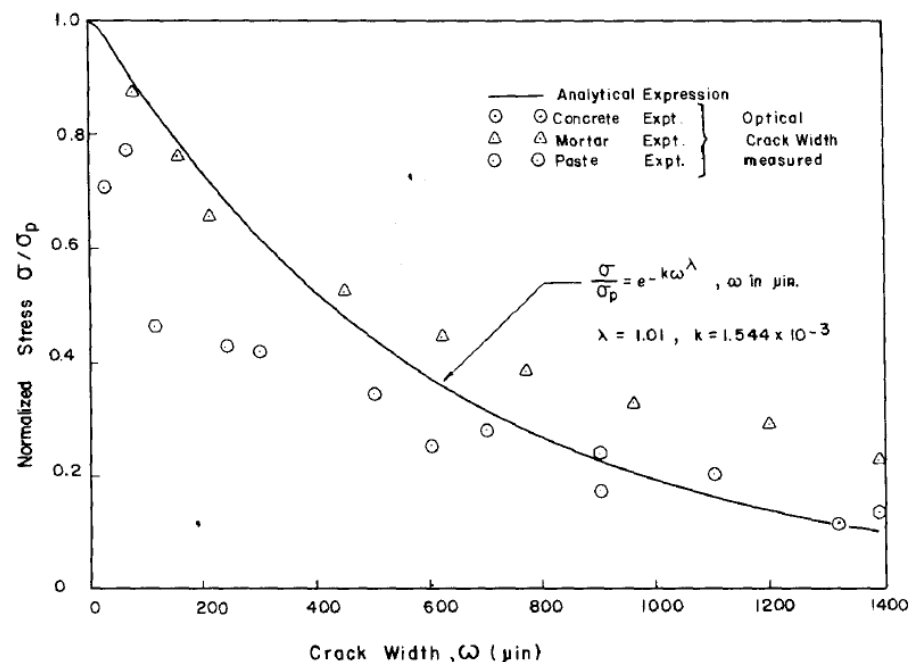
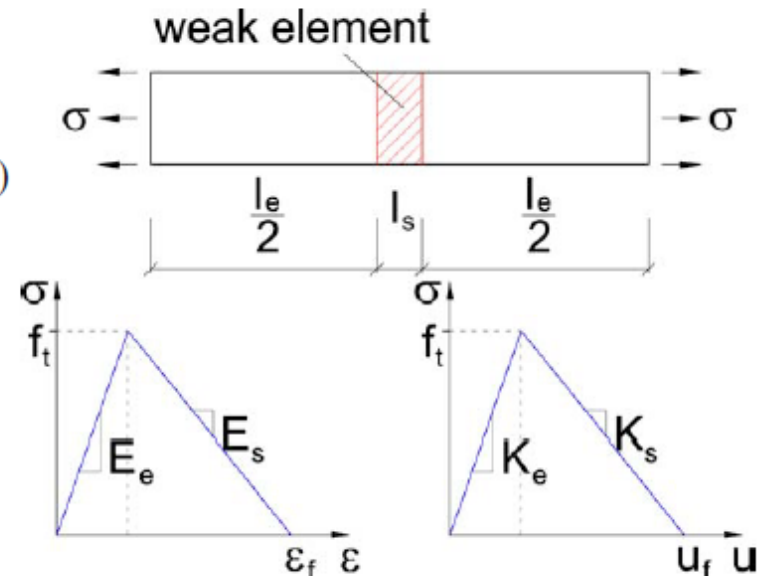


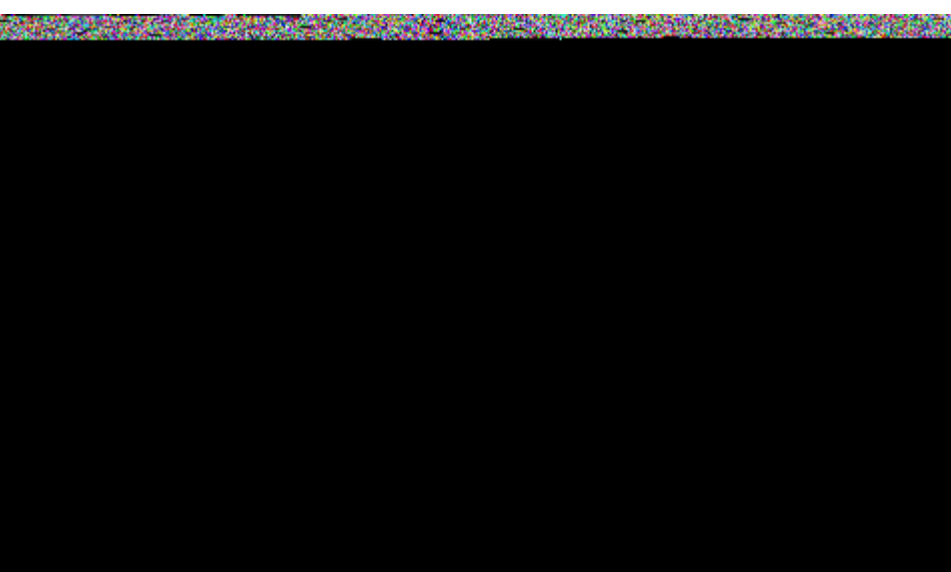
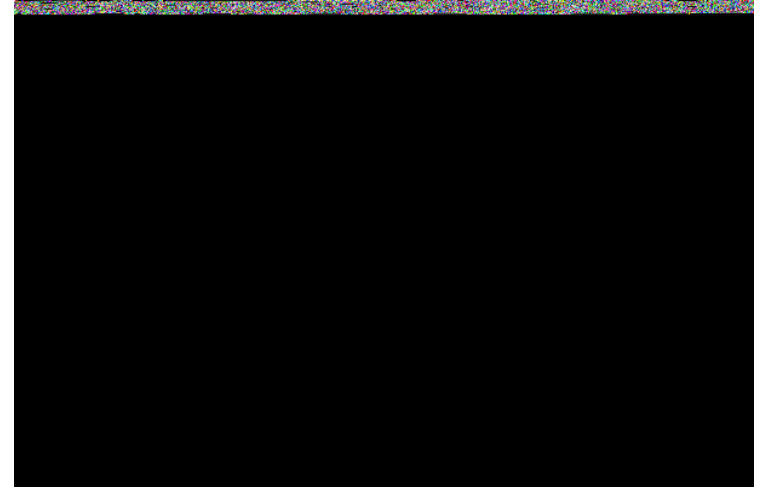
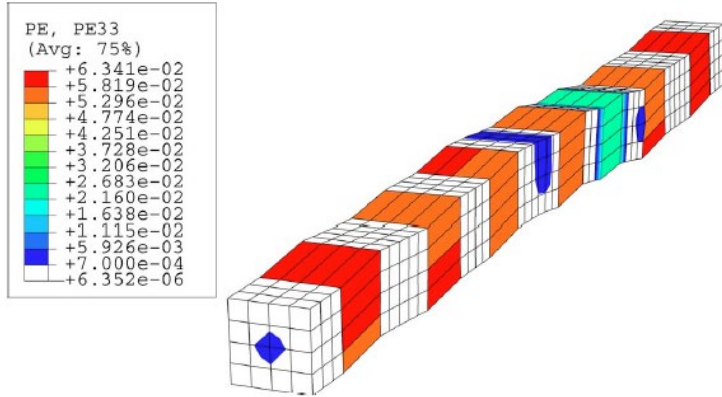
Fig. 10 — A comparison of some of the existing analytical models used to represent the softening behavior of concrete in uniaixal tension

Tension Softening, uniqueness of the problem, local vs, Global strain

$$\Delta \ell = \left[\frac{\ell_s}{E_s} + \frac{\ell_e}{E_e} \right] \Delta \sigma \quad \text{or inversely} \quad \Delta \sigma = \frac{E_s E_e}{E_e \ell_s + E_s \ell_e} \Delta \ell \quad (3)$$



Koeberl and Willam Tension Stiffening



Strain hardening & multiple cracking



Redundancies in load distribution, tension and Flexure



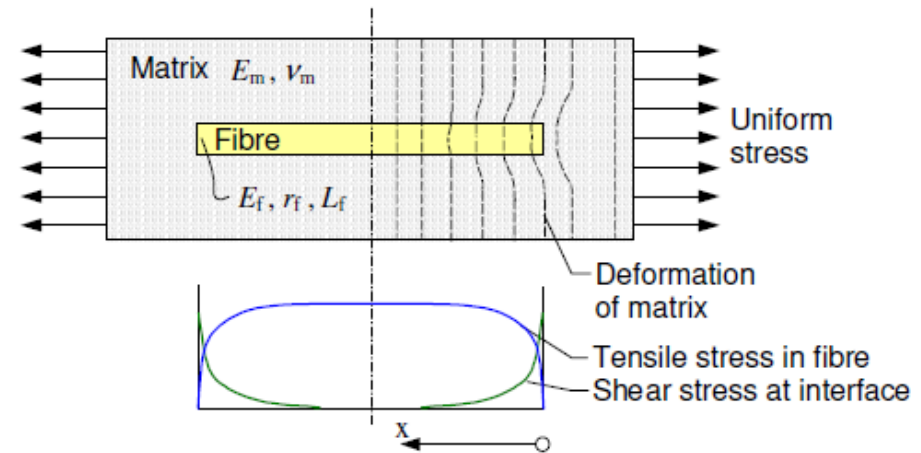
Softening
material

Hardening
material



ARIZONA STATE
UNIVERSITY

Fiber Bridging, Shear Lag, Piggott, Cox



Modeling of Stress Crack width relationship

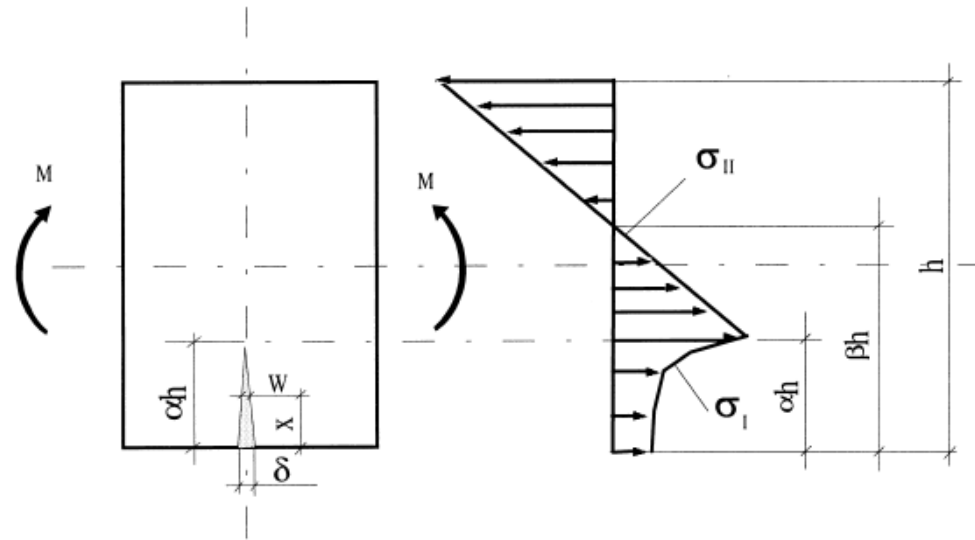
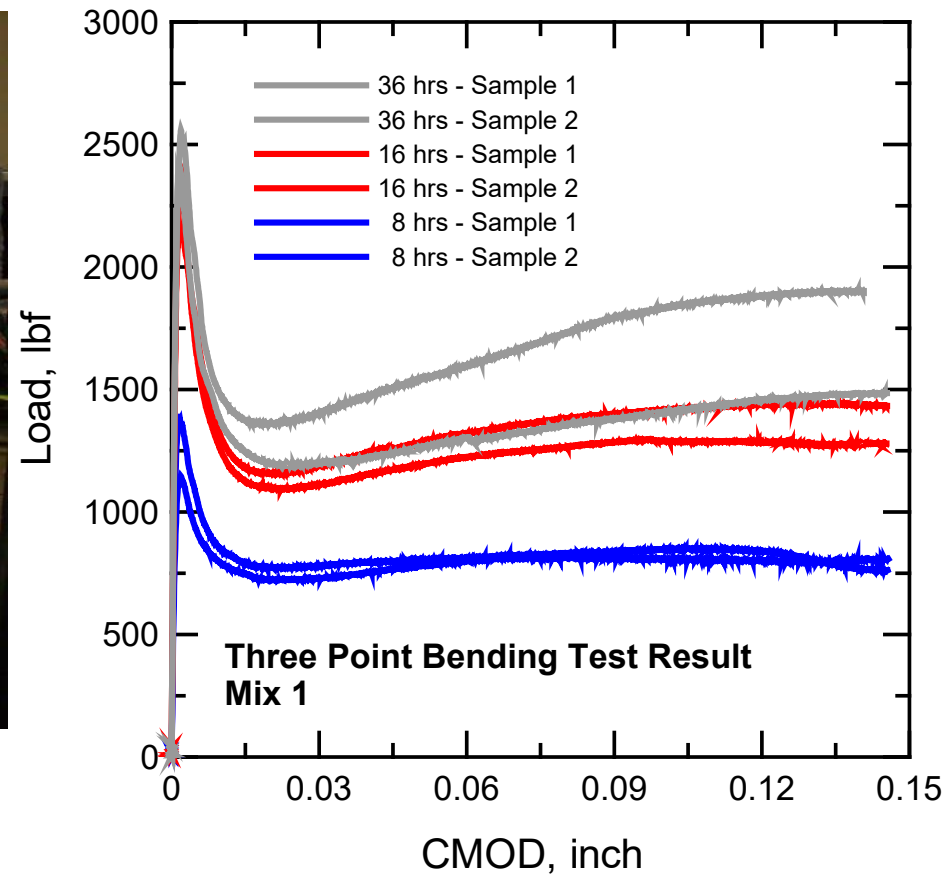


FIG. 1.

Distribution of normal stress in the cracked beam section.

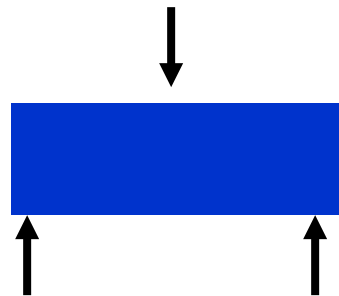
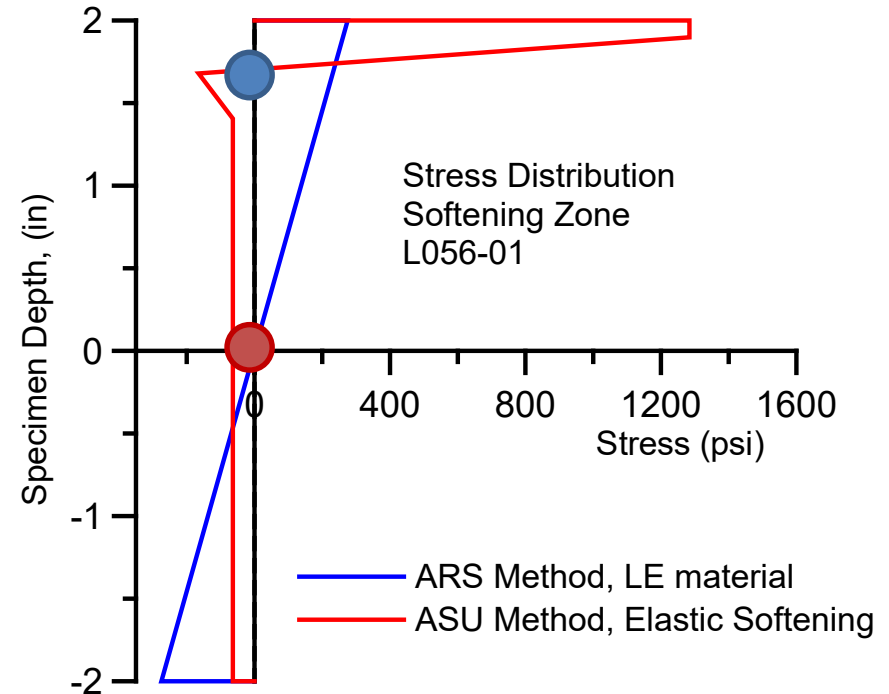
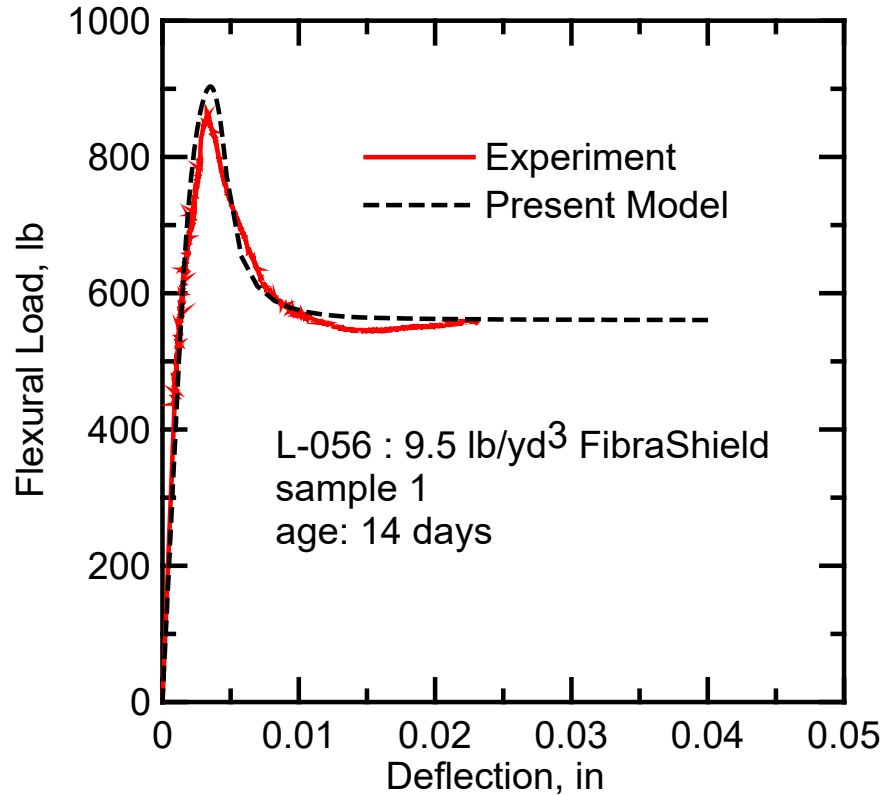
Consideration of Residual strength Effect



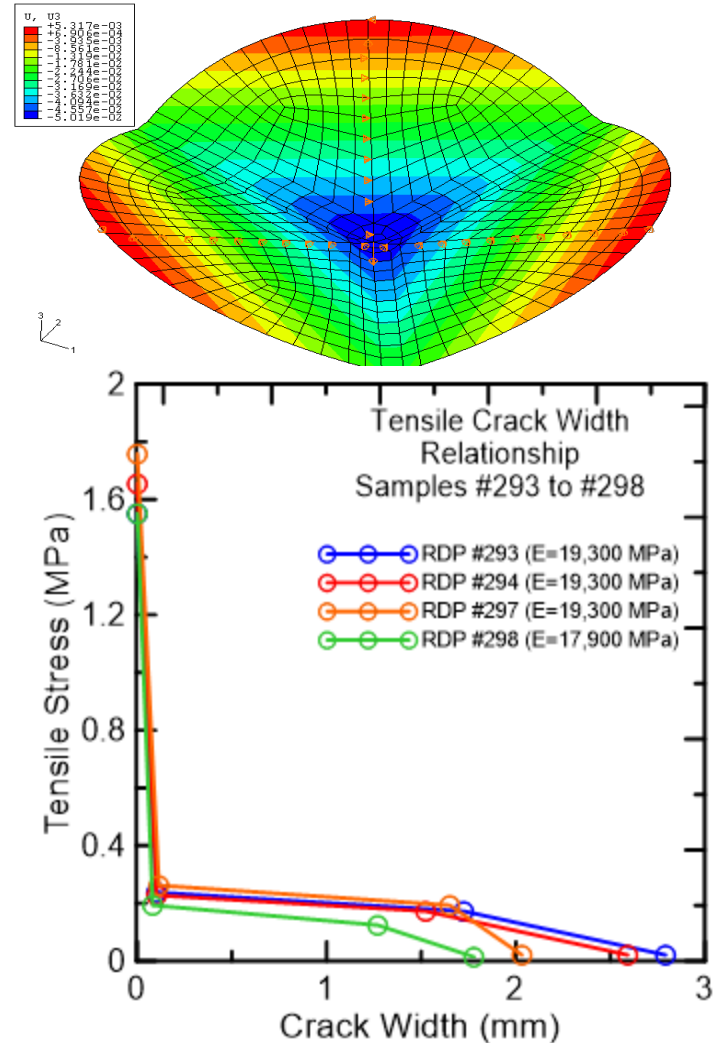
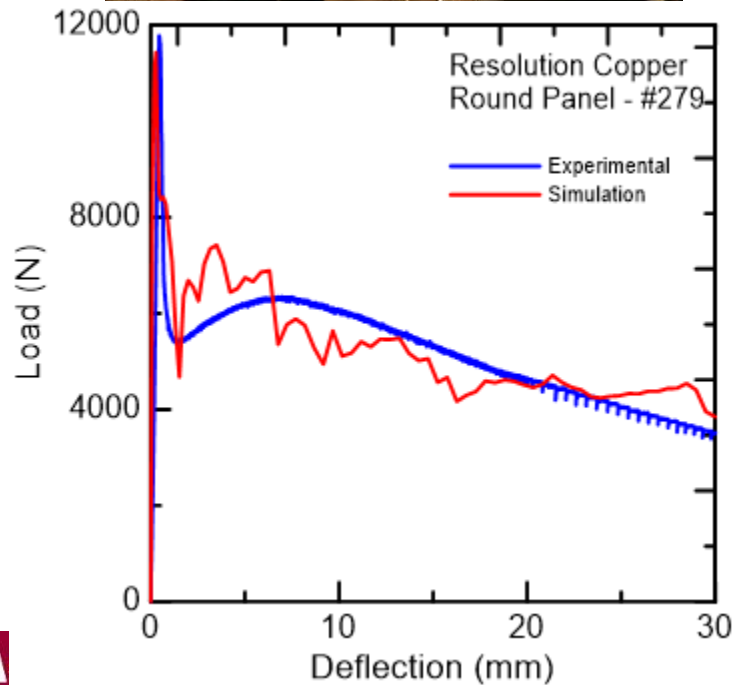
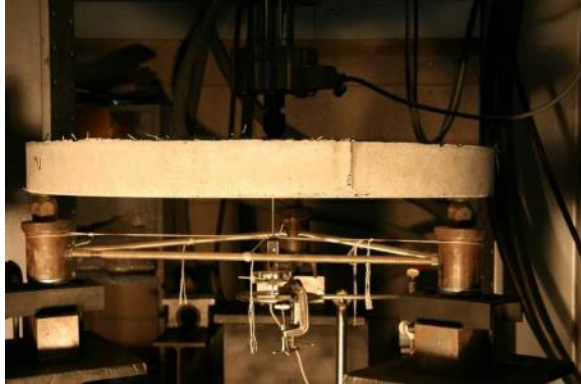
10-12 lbs/yd³ of macro fibers



Stress distribution for a cracked material with Strain-Softening FRC



ASTM C1550-Round Panel 3P-support specimen

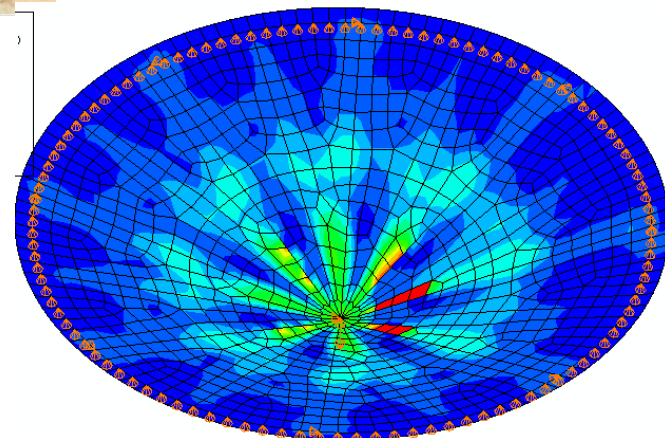
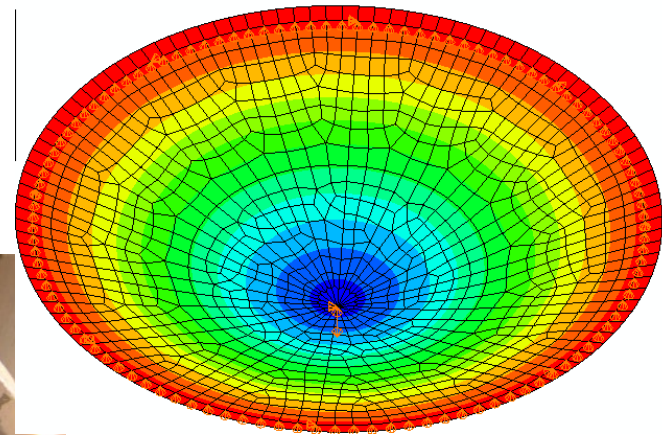
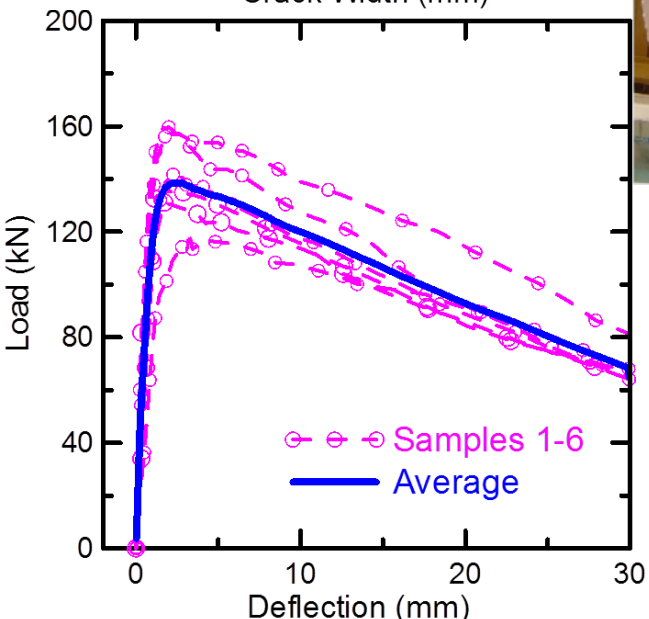
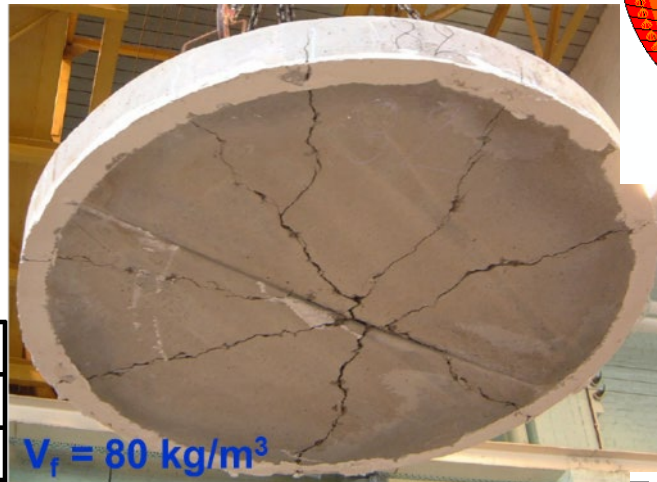
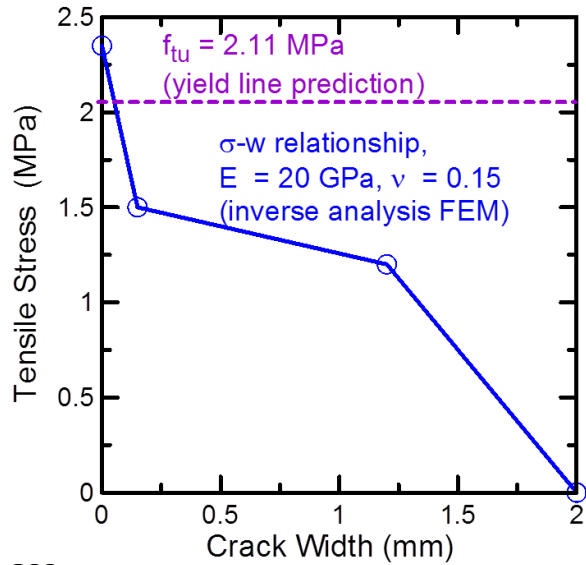


Round panel tests for evaluation of SFRC

- Test setup
 - displacement controlled, continuous support, center point load
- Dimensions
 - $D, t = 1500, 150$ mm
 - stroke diameter = 150 mm



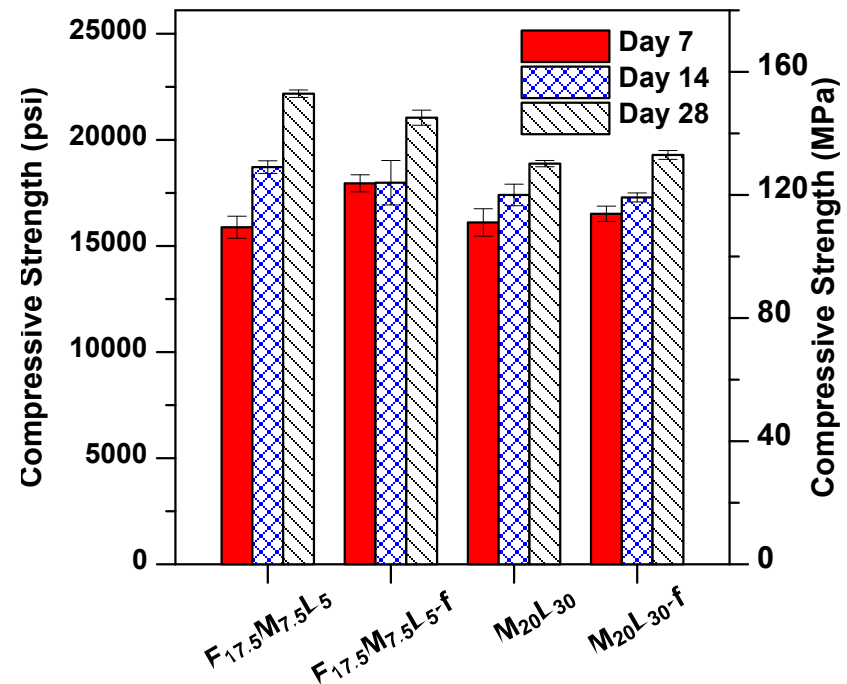
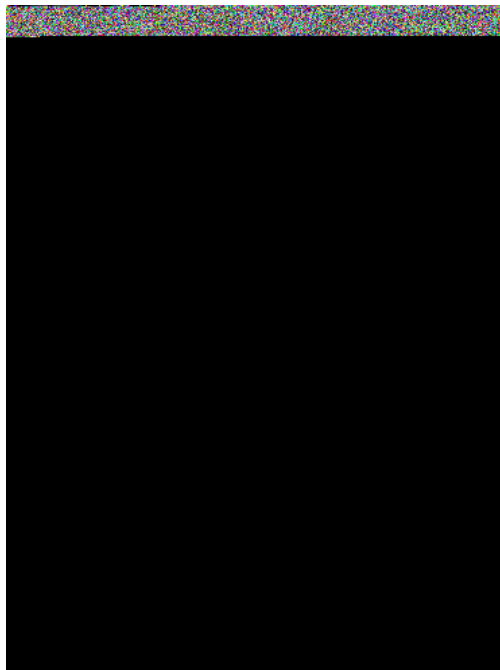
Round Panel Continuous Support Specimens





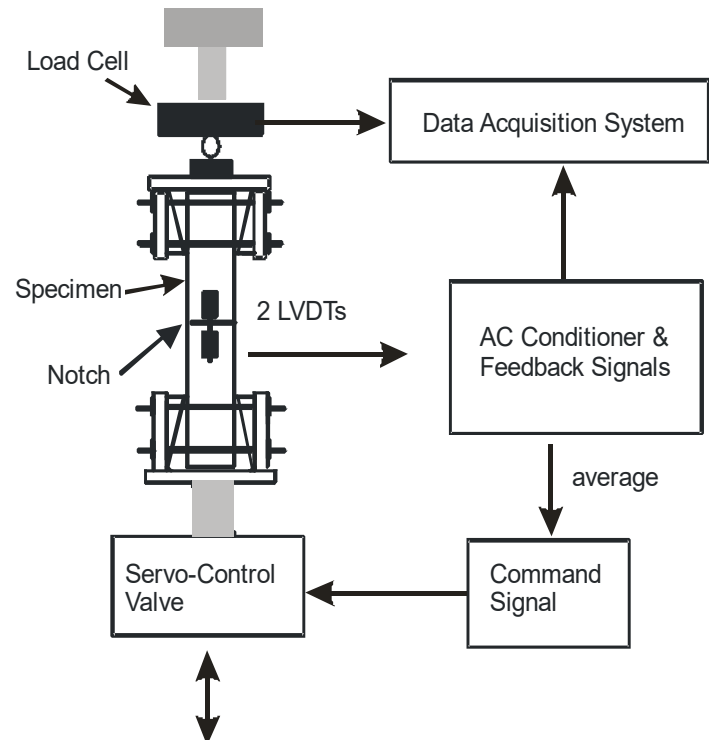
Compression Strength of UHPC Concrete Mixtures

- Compressive strength evaluated at 7,14 and 28 days of hydration provides the rate of strength gain
- 28 day strengths of at least 130 MPa were obtained for all concrete mixtures.

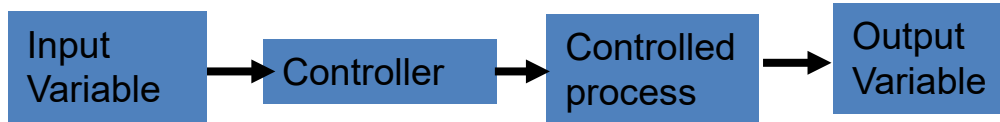


Open-Loop vs. Closed-Loop Testing

- In an open-loop test, the machine applies the load at a constant rate until failure or reaching a specified limit.
- In a closed-loop test, instrumentation is used and feedback signals control the rate of testing, e.g. with LVDT or extensometer, somehow similar to the cruise control in a car.
- Data acquisition system and signal conditioners are necessary parts of a closed-loop system. This method is used when the post-peak response is of interest.

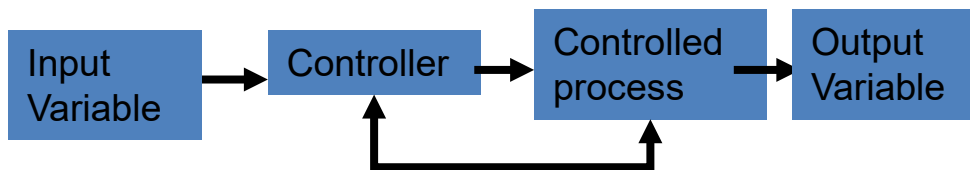
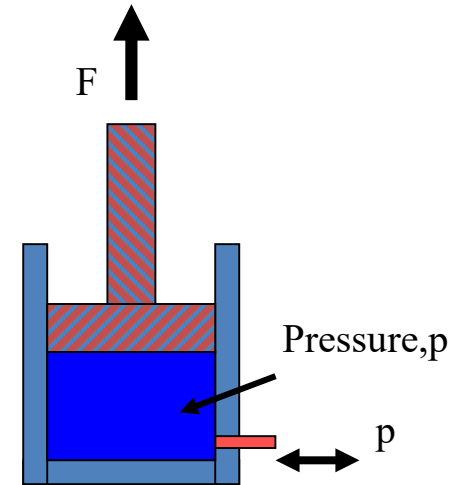


Open-Loop vs. Closed-Loop Testing



Single acting Servovalve

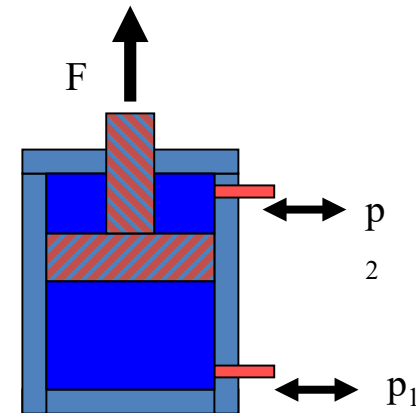
Load is proportional to pressure and can not act sufficiently fast in case of changes in the system



Feedback Signal

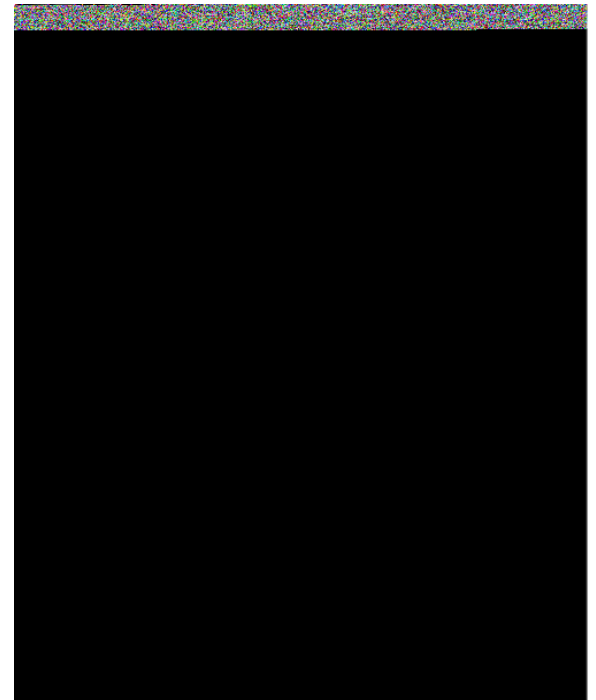
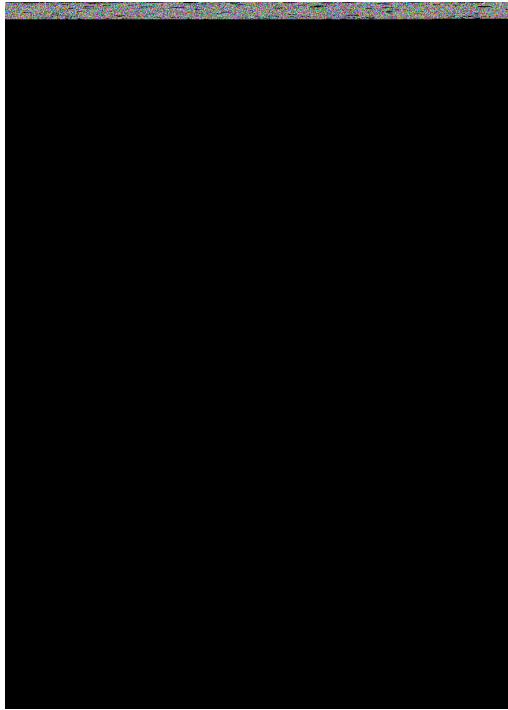
Double acting Servovalves

Load is proportional to $(p_2 - p_1)$ System can quickly respond to changes in pressure



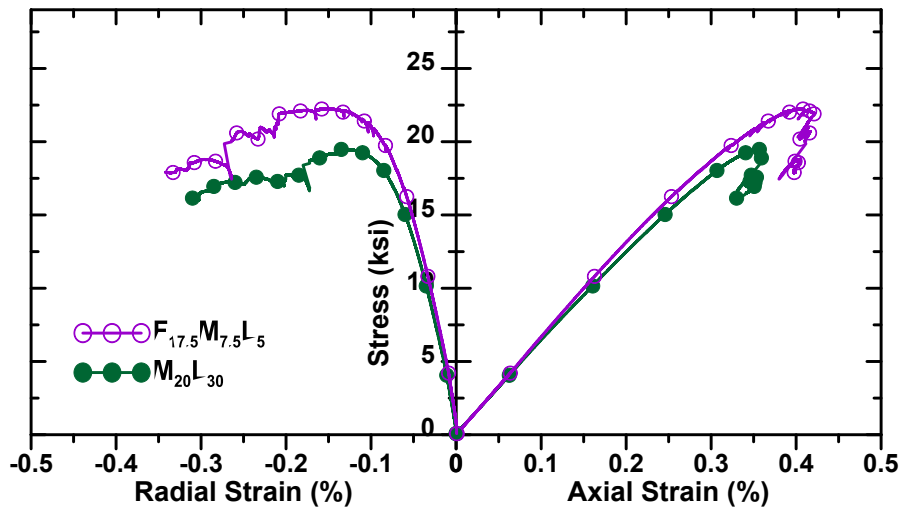
Stress-Strain Response in Compression – Test Setup

- 2" x 4" cylinders cored from 3" x 6" cylinders for evaluation of stress-strain response.
- Ends of the cylinders were ground to extremely low surface roughness (< 0.007 inches).
- In-situ ultrasonic pulse velocity (UPV) testing was also conducted during the compression test and velocity measurements were recorded at successive stress intervals of 10 MPa.

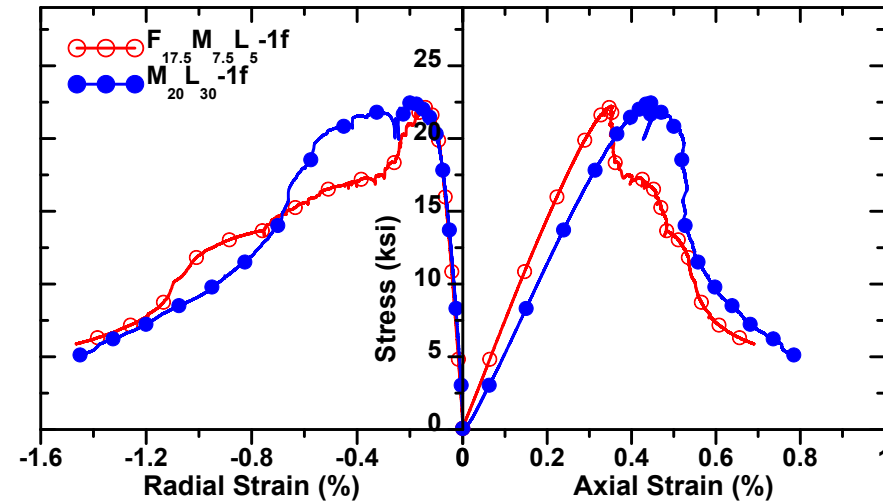


Stress-Strain Response in Compression – Effect of Mixture Composition and Fiber Volume

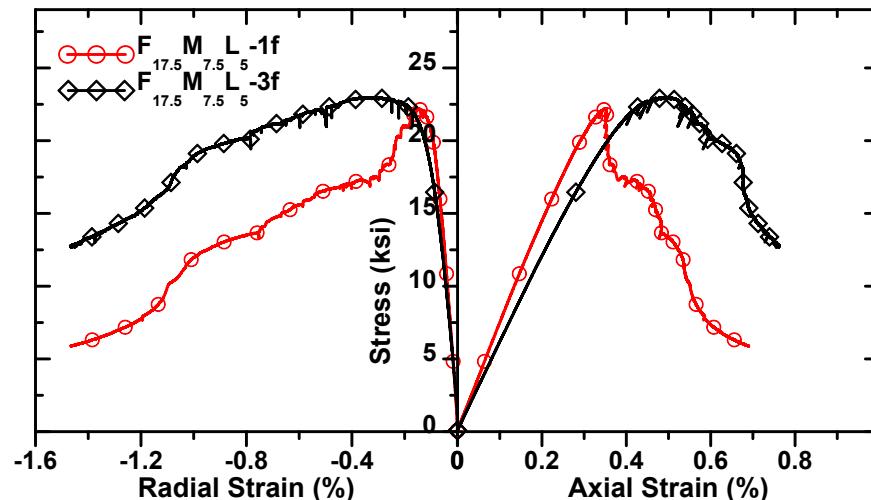
- Modulus of elasticity and Poisson's ratio can be calculated



Effect of Matrix component



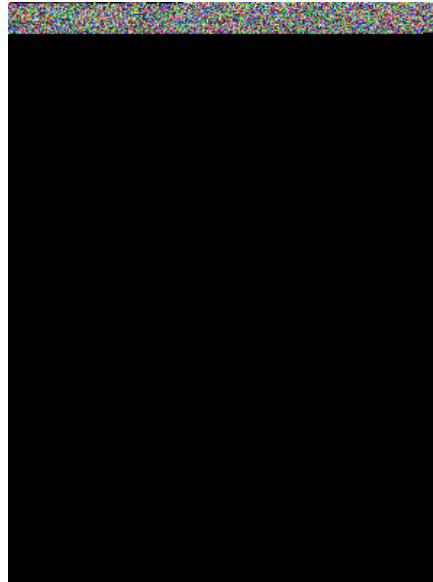
Effect of fiber on the response



Effect of fiber content

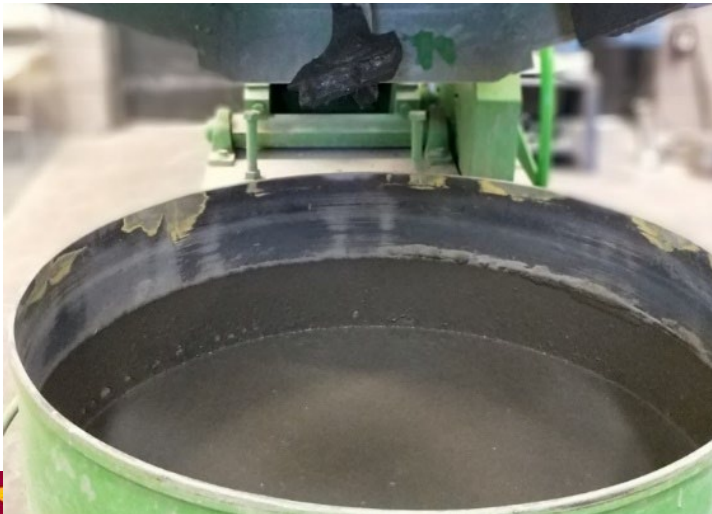
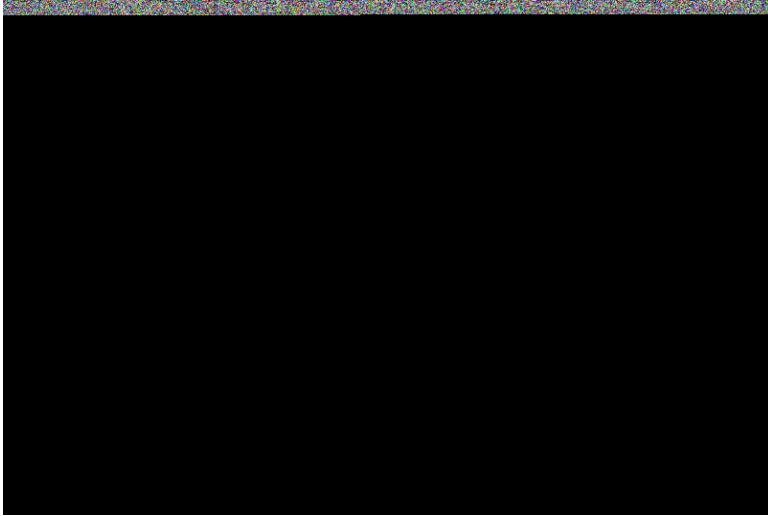
Aggregate Classes Used in UHPC

- 5 different aggregate classes were used corresponding to sizes - #4, #8, #10, (7mm and smaller), coarse sand with a $d_{50} = 0.6$ mm, fine sand with a $d_{50} = 0.2$ mm
- Steel fibers – $d = 0.6$ mm, $l = 13$ mm.



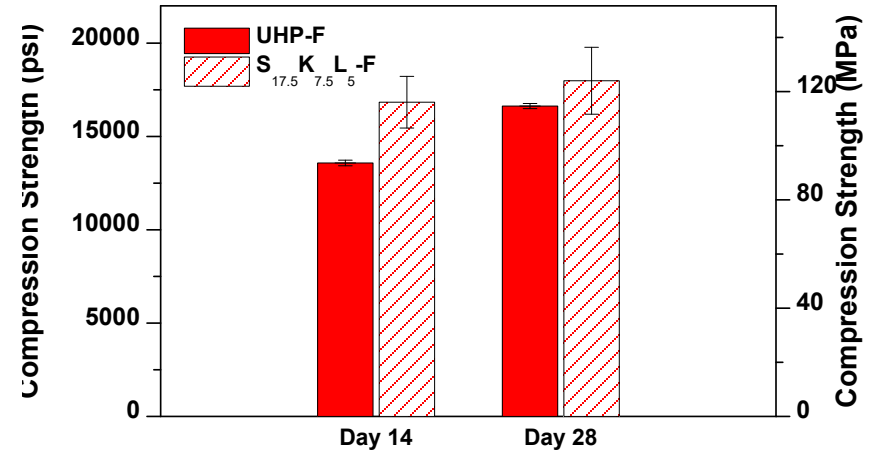
Mechanical Splitter used to obtain uniform gradation of particles

Implementation in a Large Mixer

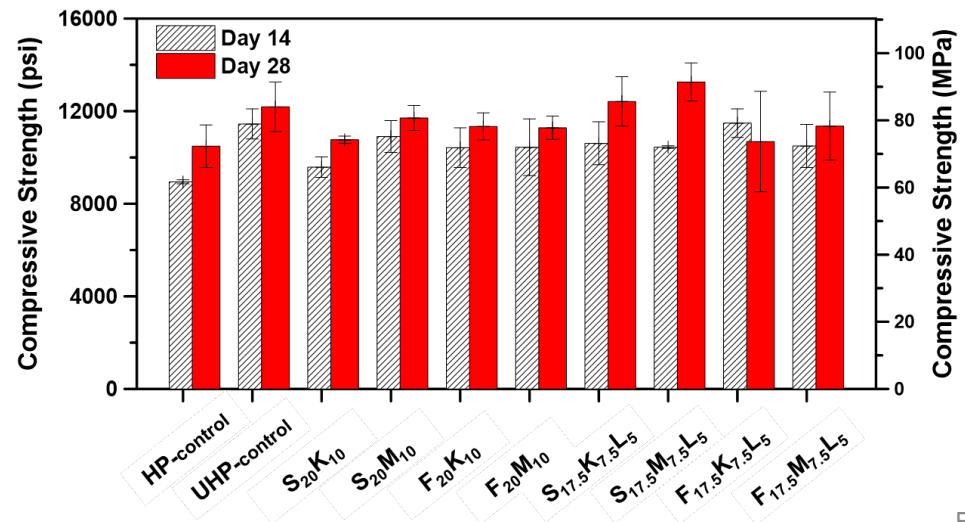


Characterization of Selected Mixtures

- based on cumulative particle reactivity (degree of hydration)
- durability metrics based on porosity and pore structure
- based on compressive strength

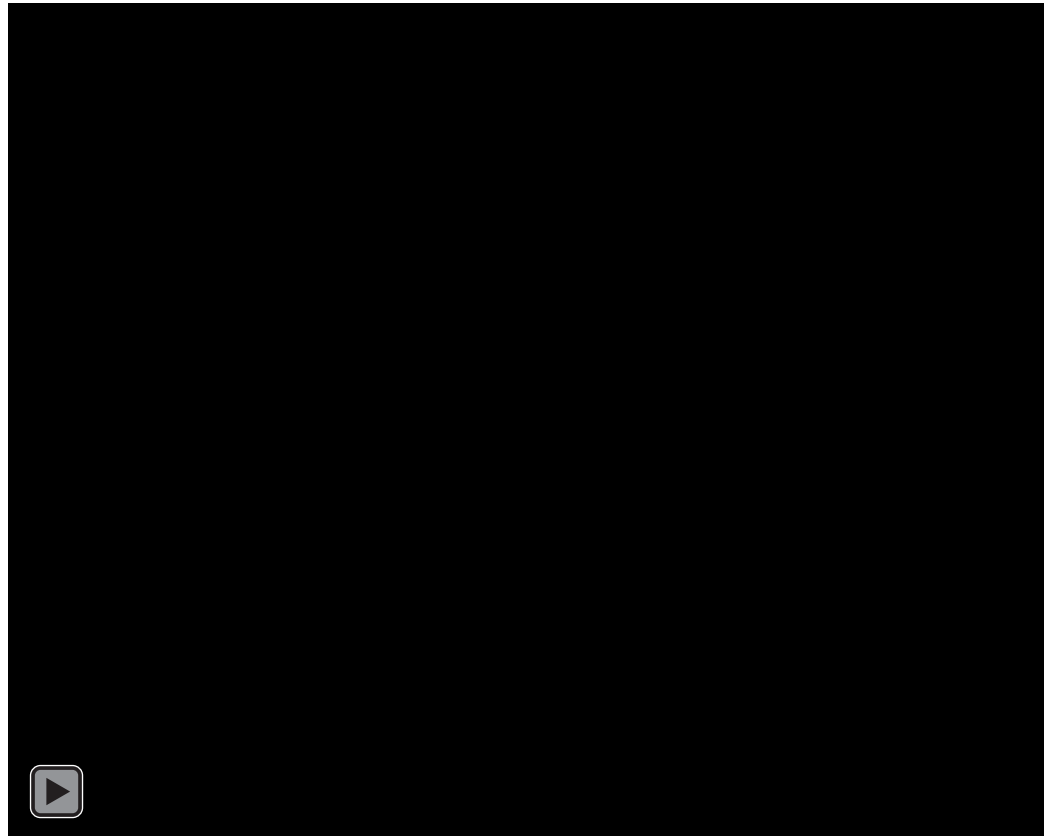


- Incorporation of steel fibers resulted in a 28 Day compressive strength of ~125 MPa.



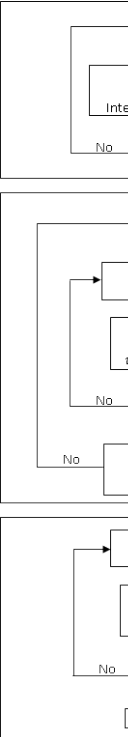
- 28 day strengths comparable to, or higher than control, even with a clinker factor reduction of 30%

Why do cracks weaken a material?

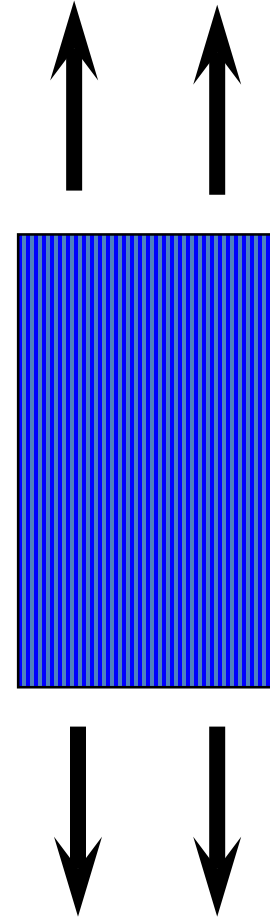
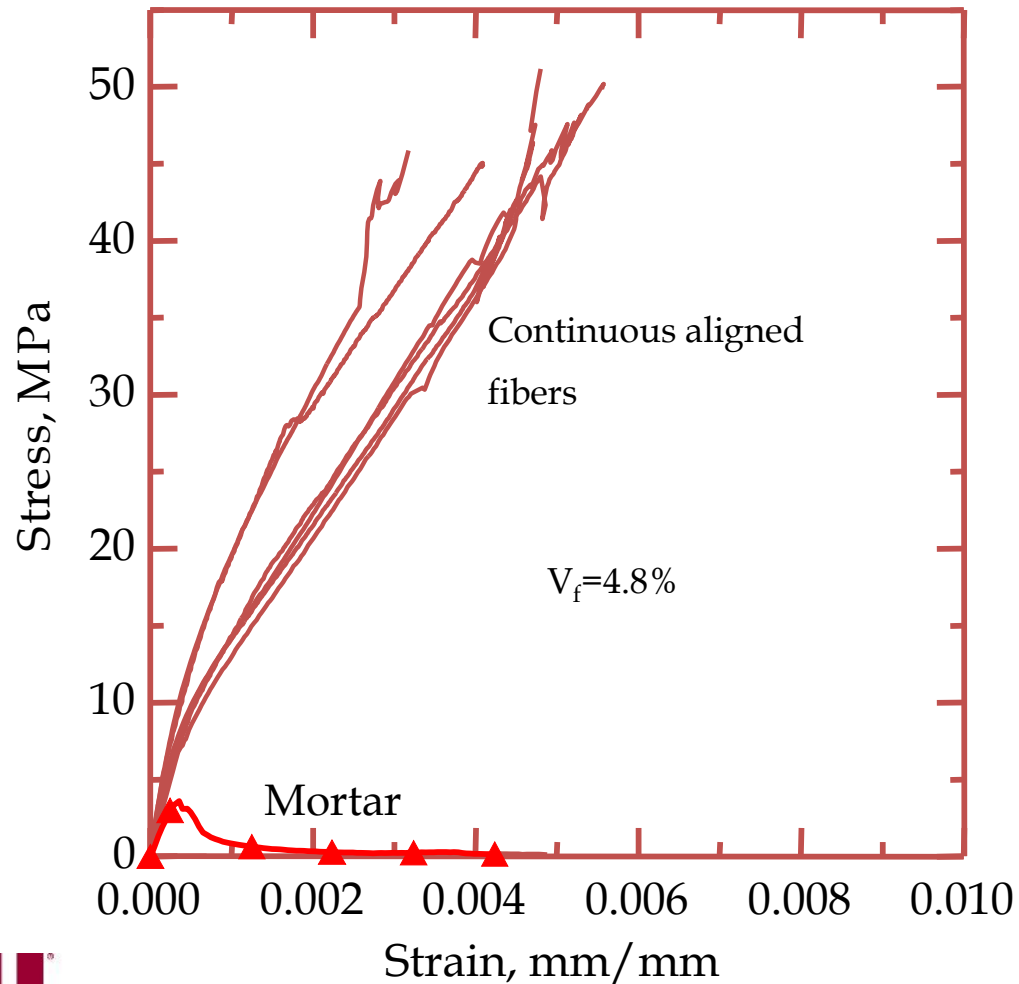


http://www.doitpoms.ac.uk/tlplib/brittle_fracture/sub_critical.php

https://www.doitpoms.ac.uk/tlplib/brittle_fracture/videos/paper_fracture.mp4

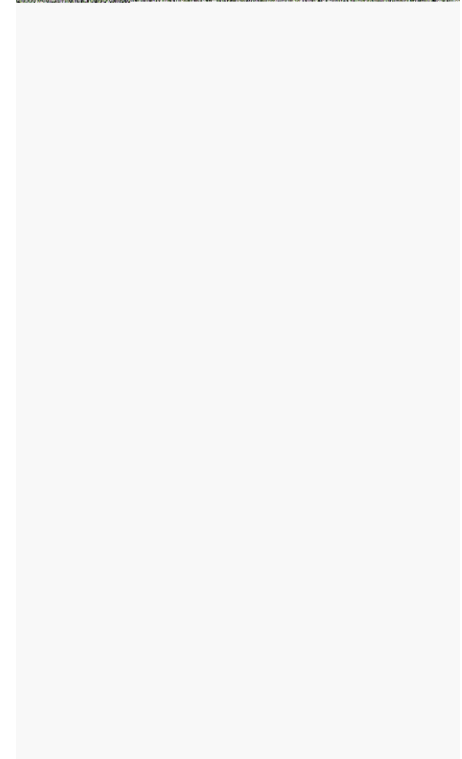
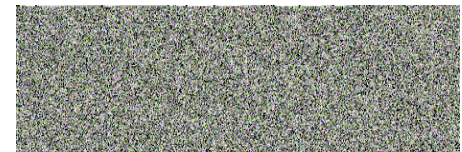
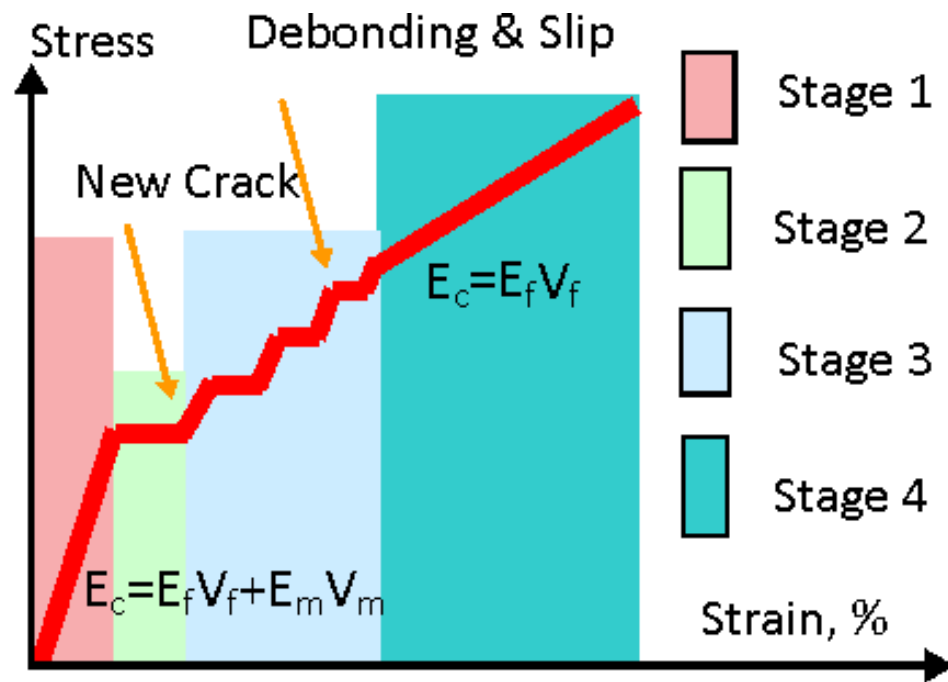


FRC materials with 50 MPa Tensile Strength

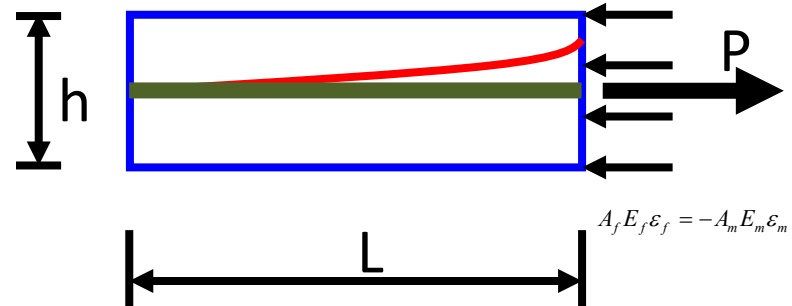
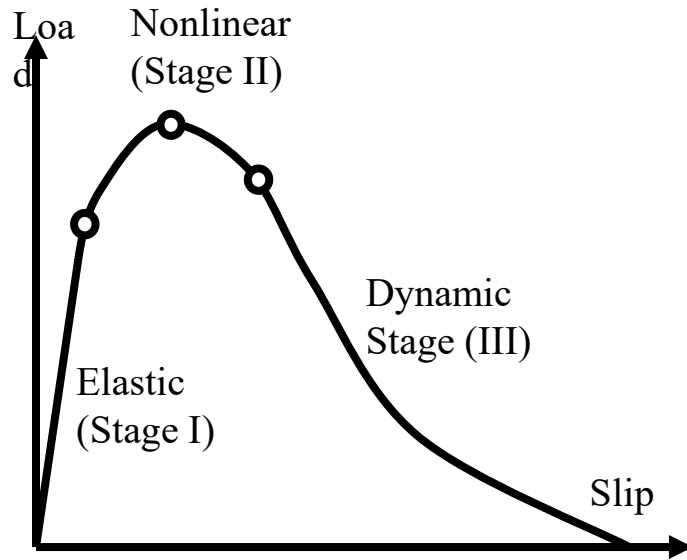


A look of sequence of microcrack formation

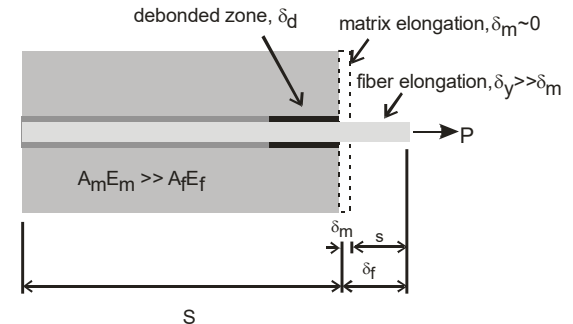
- Two types of cracking:
 - Minimal cracking in the cement paste which are perpendicular to the fiber orientation.
 - Slip of the fiber bundles after the matrix cracking.



Pullout Load Curve and Shear Strength Parameters

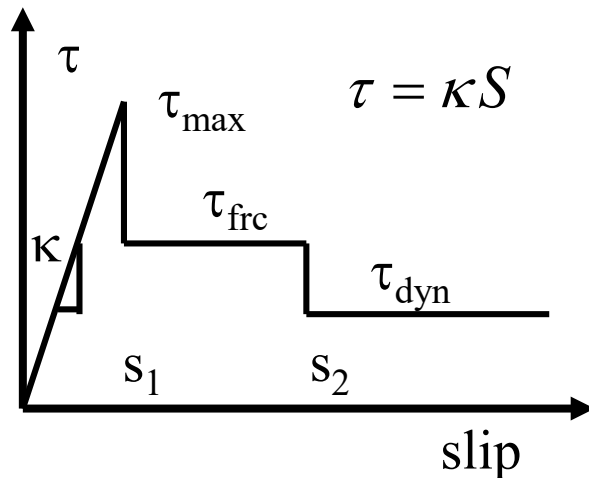


$$\frac{dP}{dx} = -\frac{dM}{dx} \psi \tau$$



$$\kappa = \beta G$$

$$G = \frac{E_m}{2(1+\nu)}$$

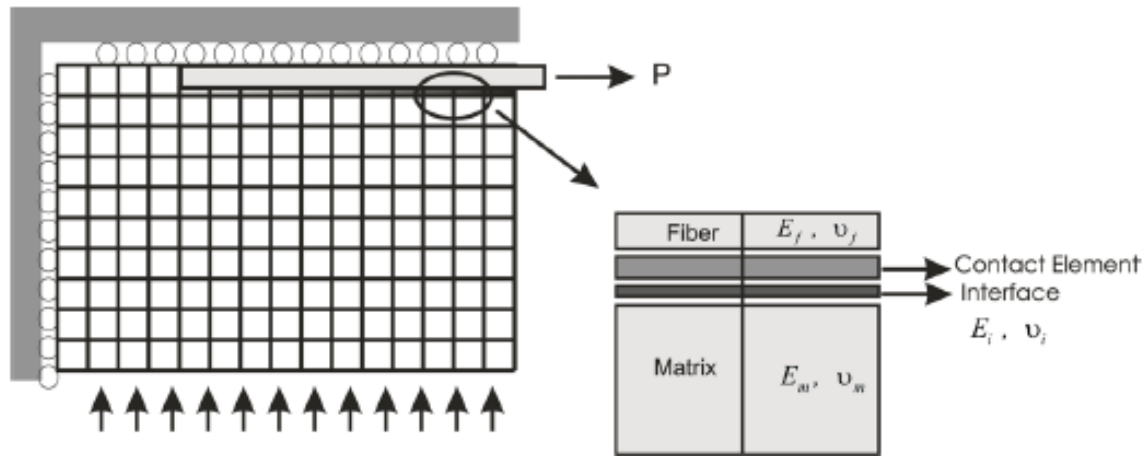


$$S = (\delta_f - \delta_m) = \int_0^x [\epsilon_f(x) - \epsilon_m(x)] dx$$

$$\frac{d^2 P}{dx^2} = \psi \kappa [\epsilon_f(x) - \epsilon_m(x)]$$



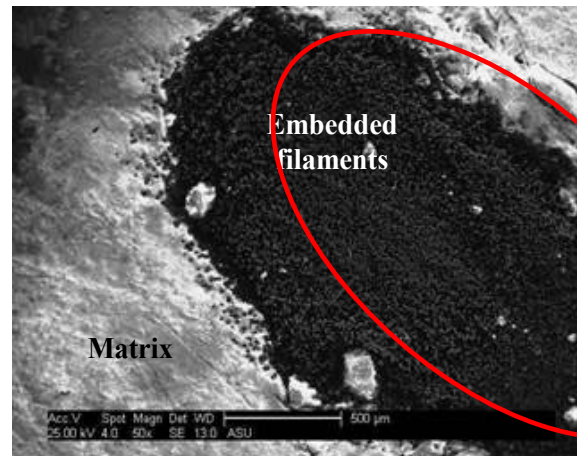
Finite Element Modeling



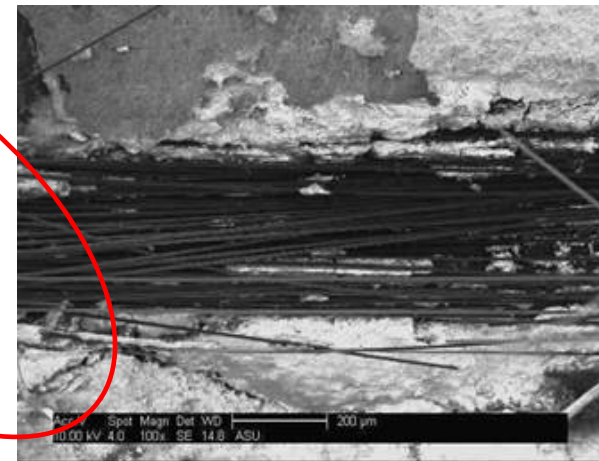
Evaluation of Microstructure

- Mixtures containing flyash result in lower viscosity and better flow of the matrix around the surface of the fabric.
- Interstitial fabric filling by flyash particles, Reduction in capillary voids and the transition zone in the vicinity of the inter yarn spacing
- Strong bonding in mixtures with high levels of flyash.

Bundle with coating



(a)



(b)

AR glass fabric embedded in cement matrix, side view

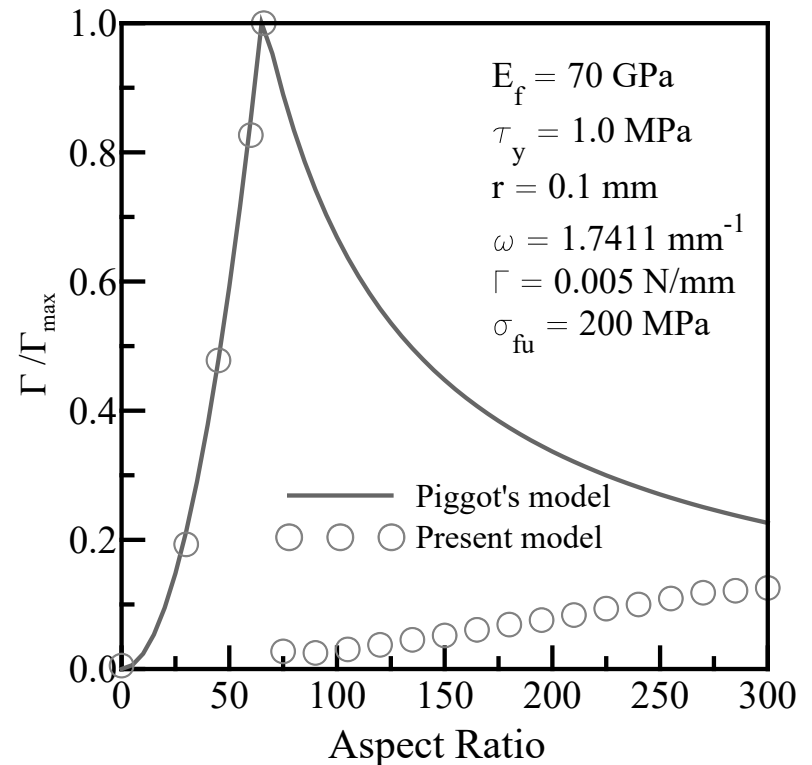
SEM micrographs of carbon fiber bundle embedded in cement matrix: (a) view of cross section and (b) side view

Energy Dissipation During Pullout

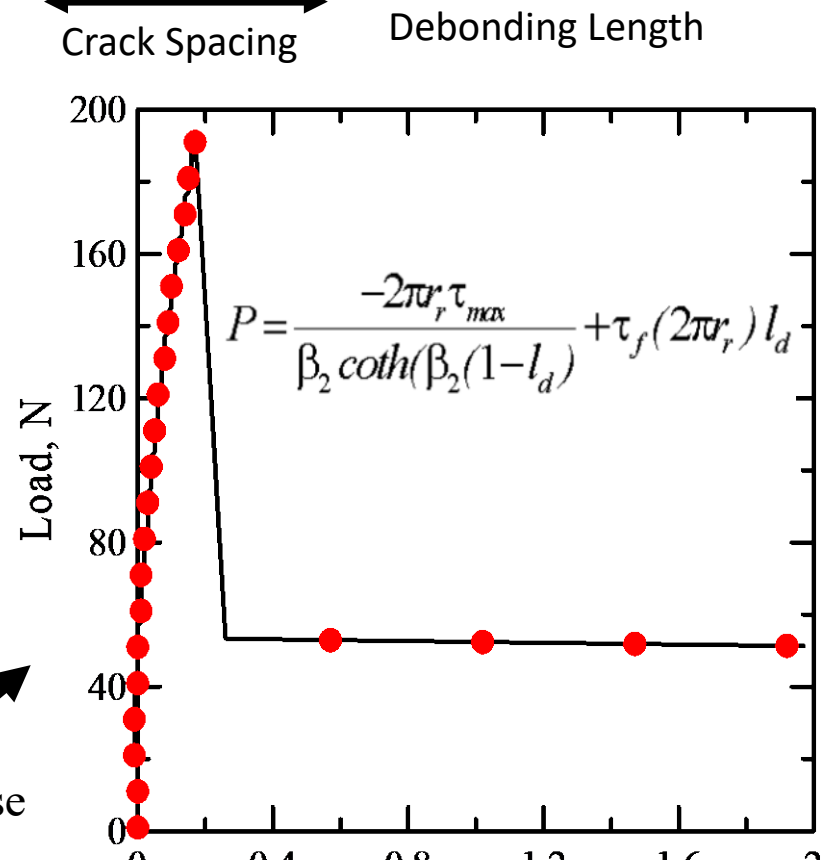
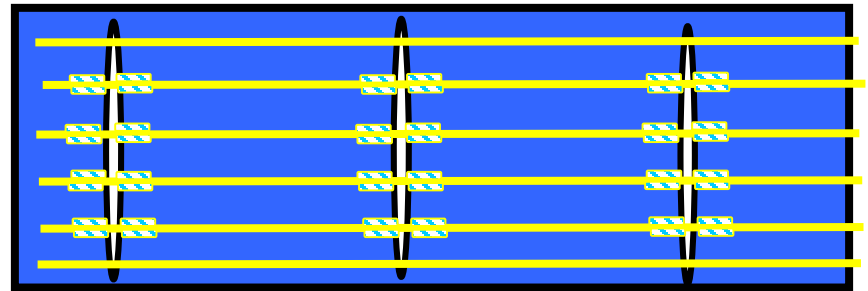
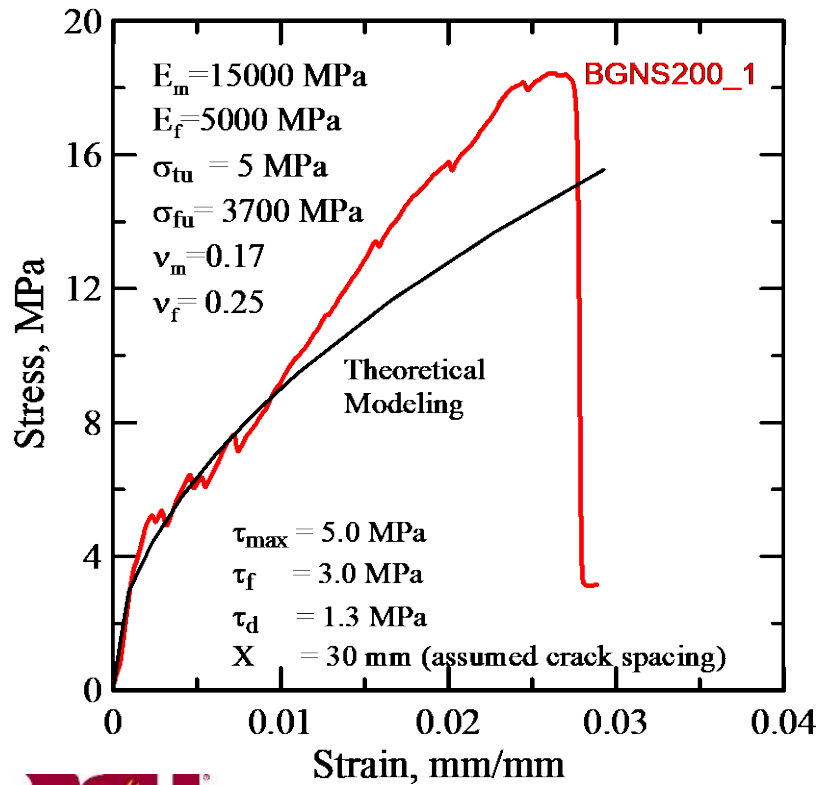
- Ductility of short fiber vs. long fiber systems as a function of fiber aspect ratio
- The load-slip response of the fiber can be used to compute γ_p as a measure of energy absorption during the debonding and pull-out process.

$$\gamma_P = \int_0^{l_p} P^*(a) \frac{du}{da} da$$

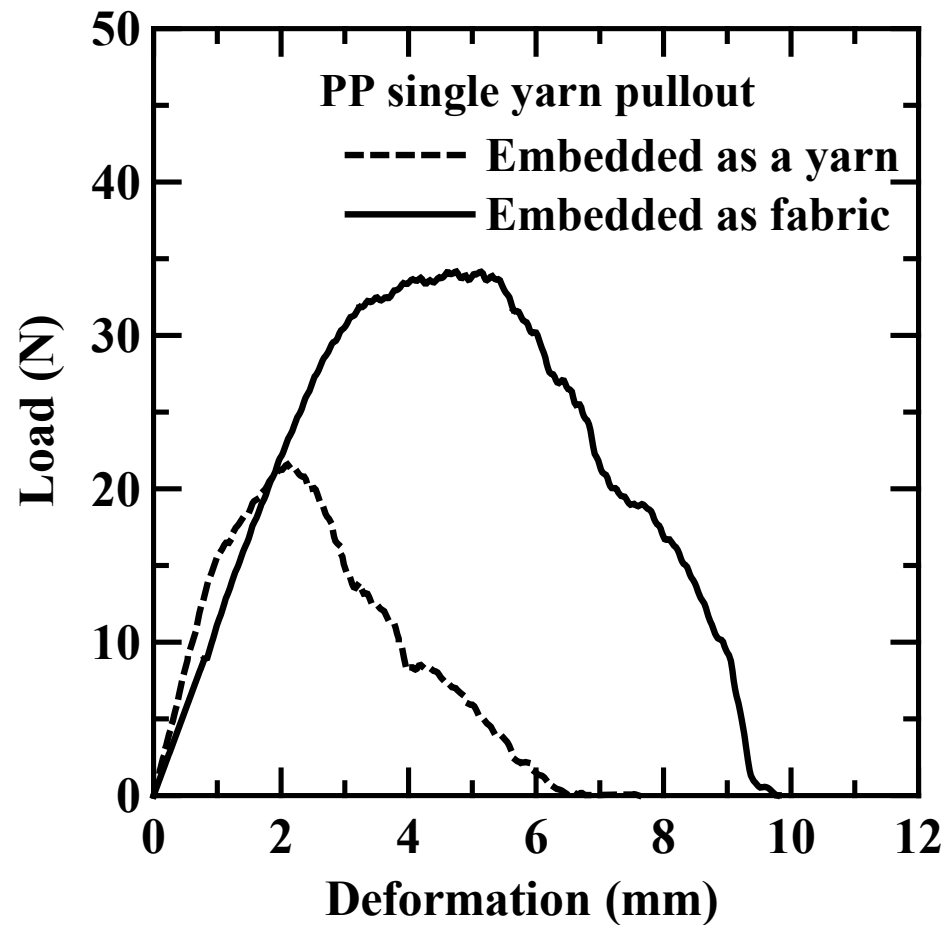
$$G_i = \frac{V_f d \sigma_{fu}^3}{6 \tau_i} \left[\frac{1}{8 \tau_i s} + \frac{1}{E_f} \left(1 - \frac{\sigma_{fu}}{2 \tau_i s} \right) \right]$$



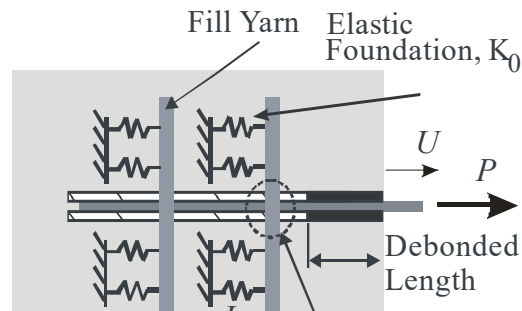
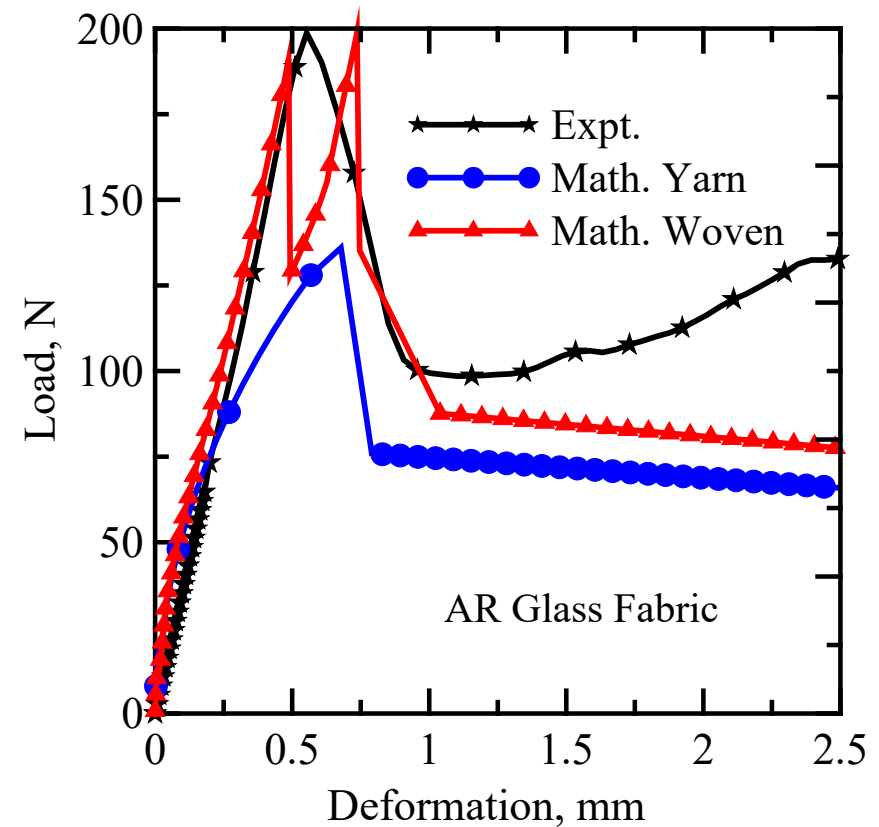
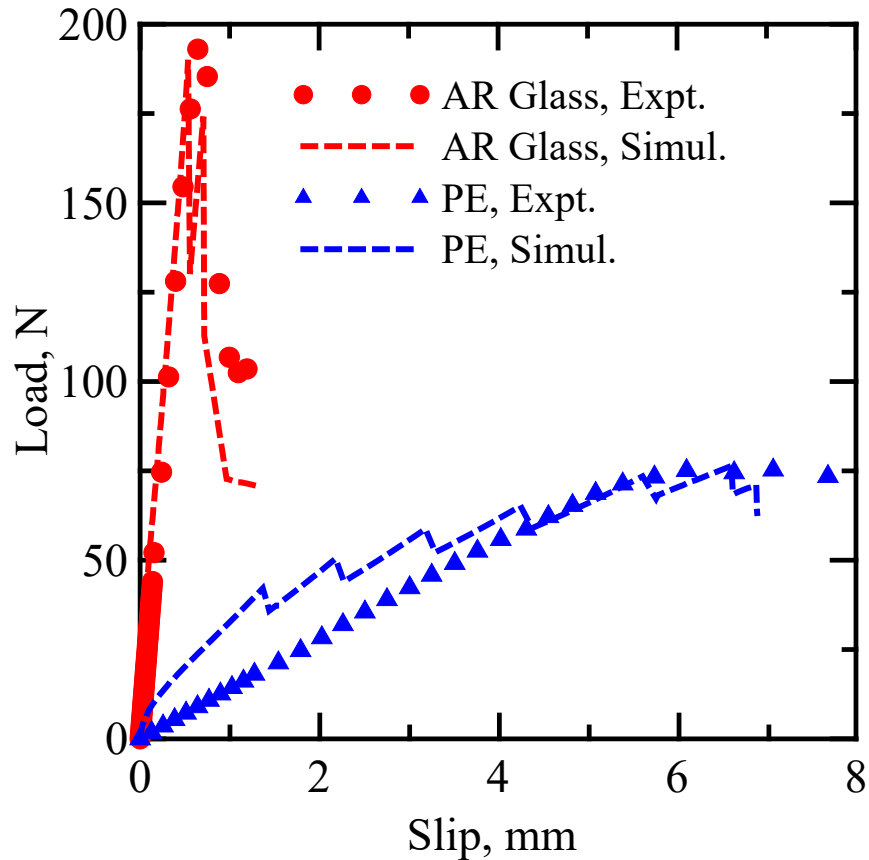
Modeling of delamination in textiles



Effect of fill yarn Anchorage in textiles



Modeling of Fabric Pullout



Durability

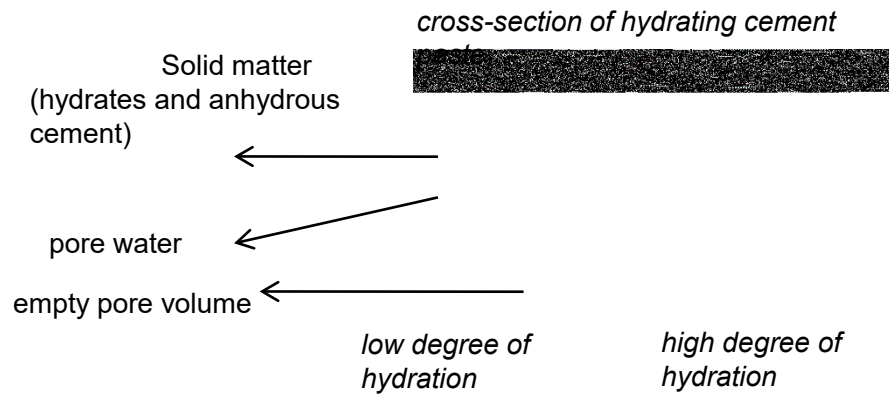
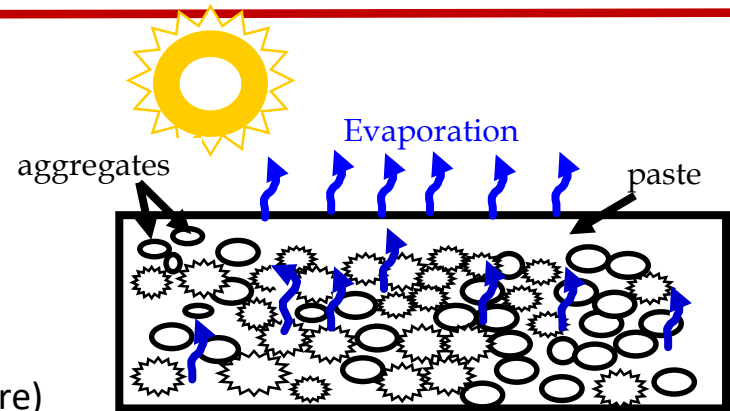
- Why the durability matters. Shrinkage effects, creep deformation, fatigue loading

Shrinkage and creep

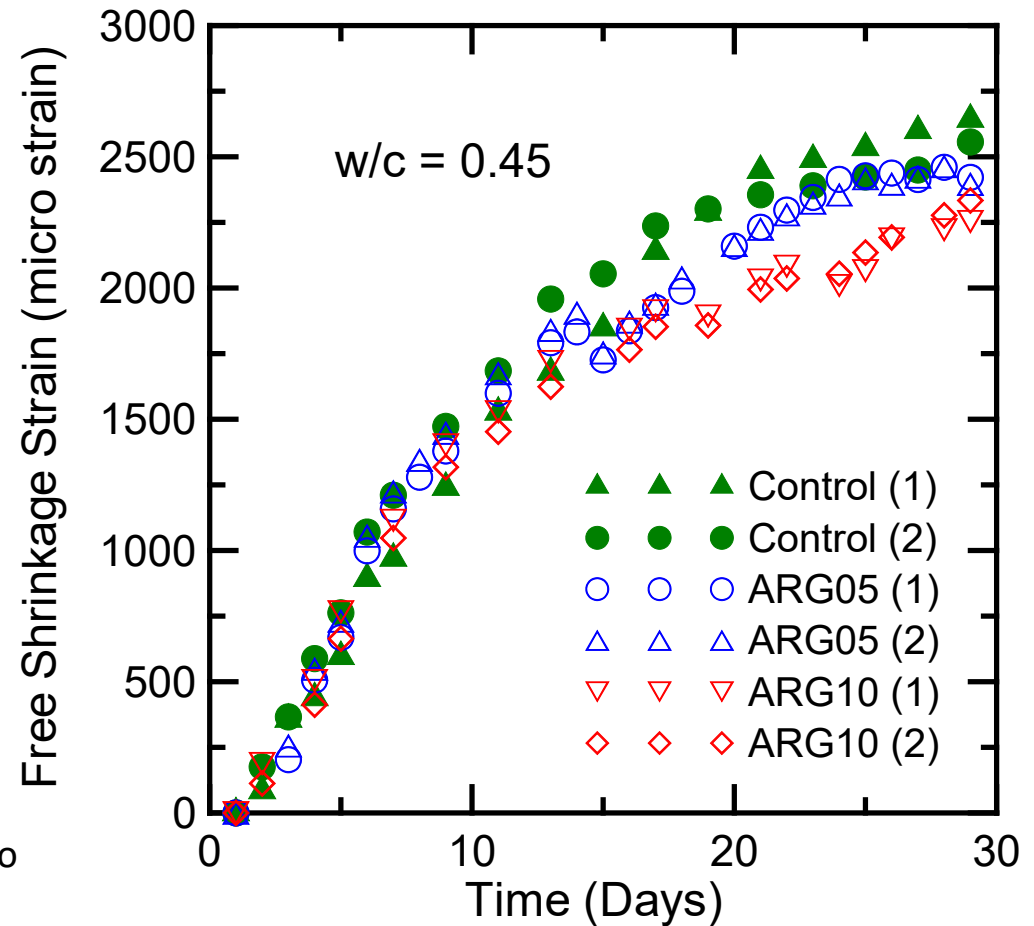
- Why is FRC superior in terms of shrinkage and creep

Shrinkage as a Volumetric Change

- Drying and plastic shrinkage
(due to water loss by evaporation)
- Autogenous/self-desiccation shrinkage
(due to cement hydration)
 - Thermal shrinkage (due to decrease in temperature)
- Carbonation shrinkage
(reaction of hydrated cement w/carbon dioxide)



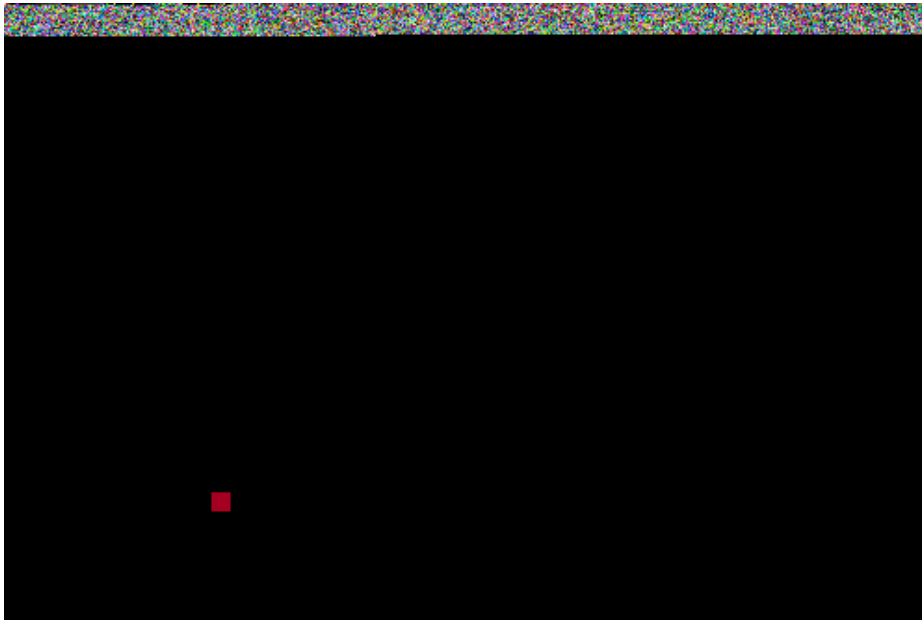
Free shrinkage test for hardened concrete



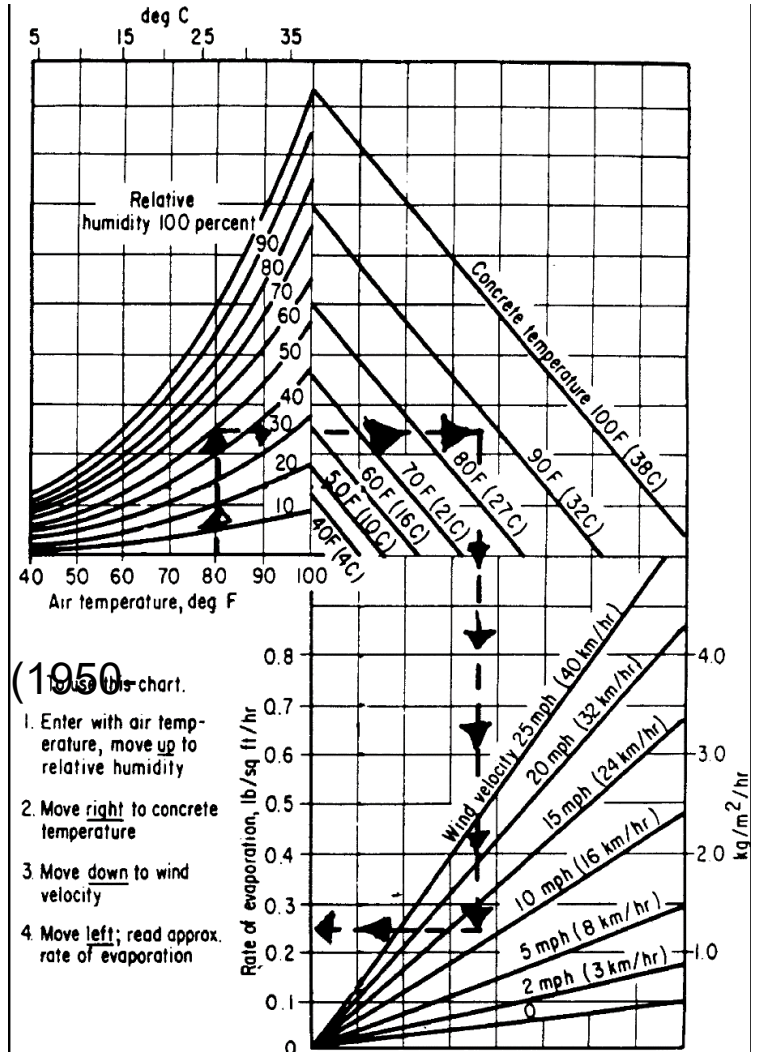
ACI nomograph for estimating rate of evaporation

- MENZEL'S FORMULA**

$$E = 0.44 (e_o - e_a)(0.253 + 0.096V)$$



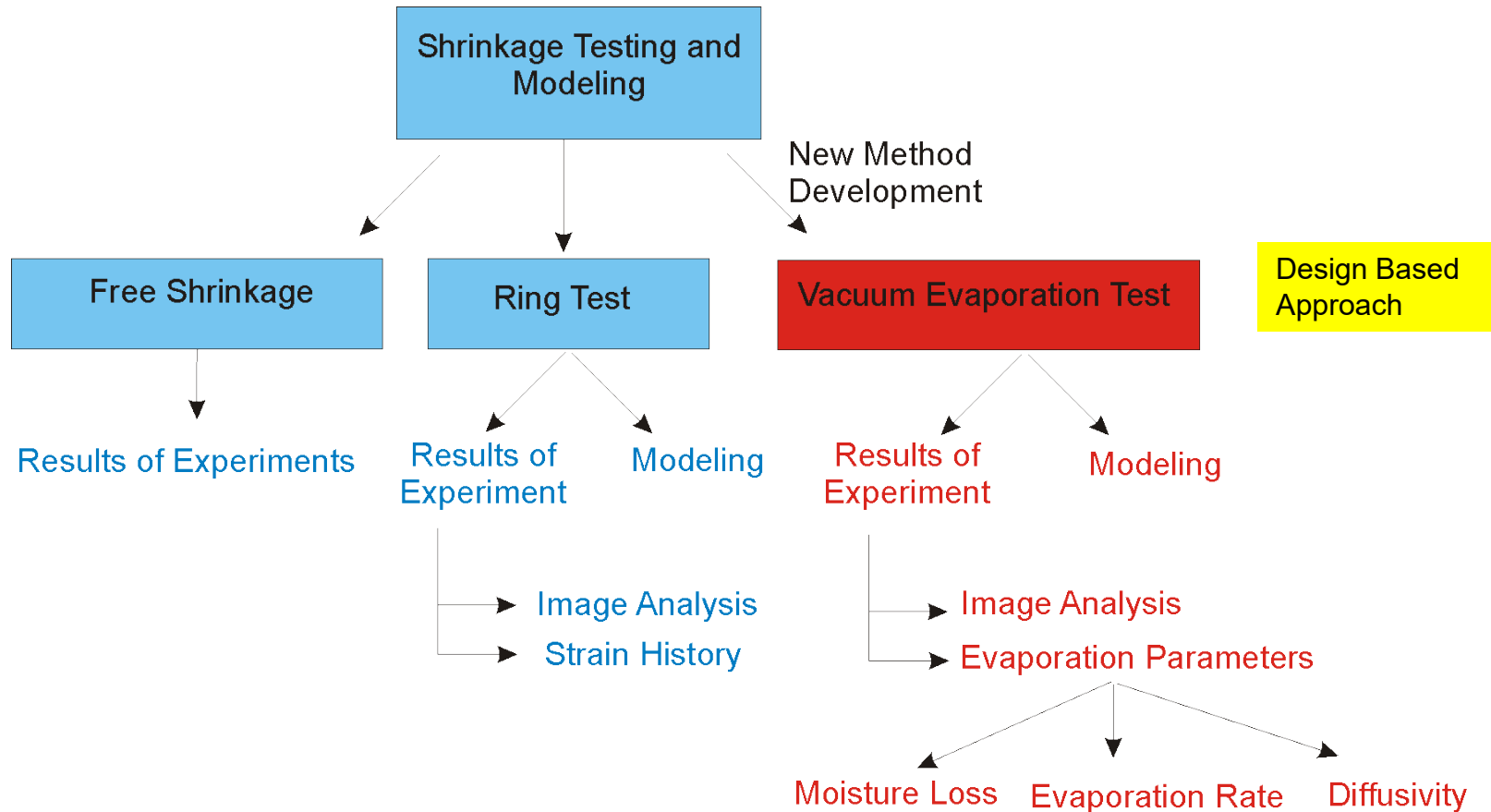
er (1950)



To Prevent Plastic Shrinkage Cracks:

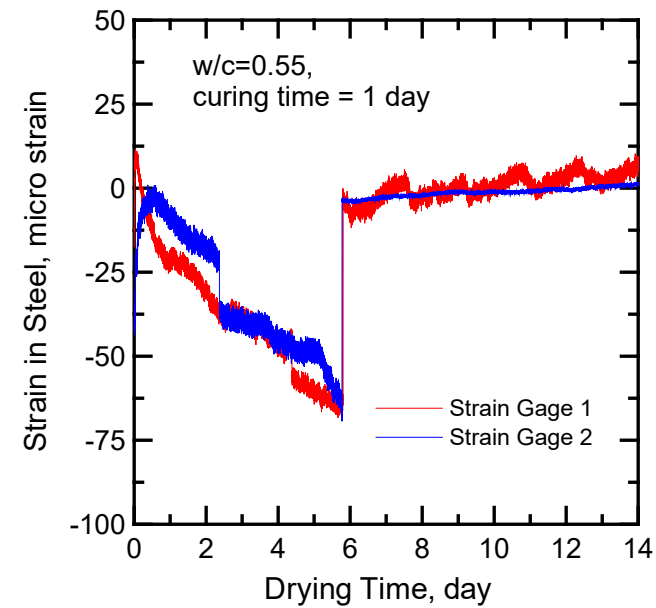
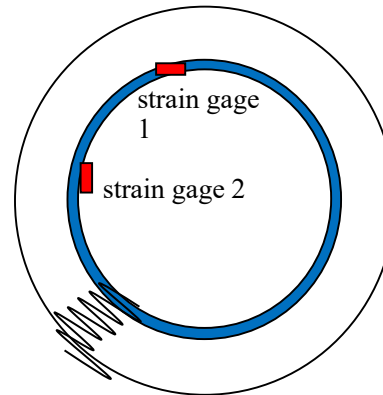
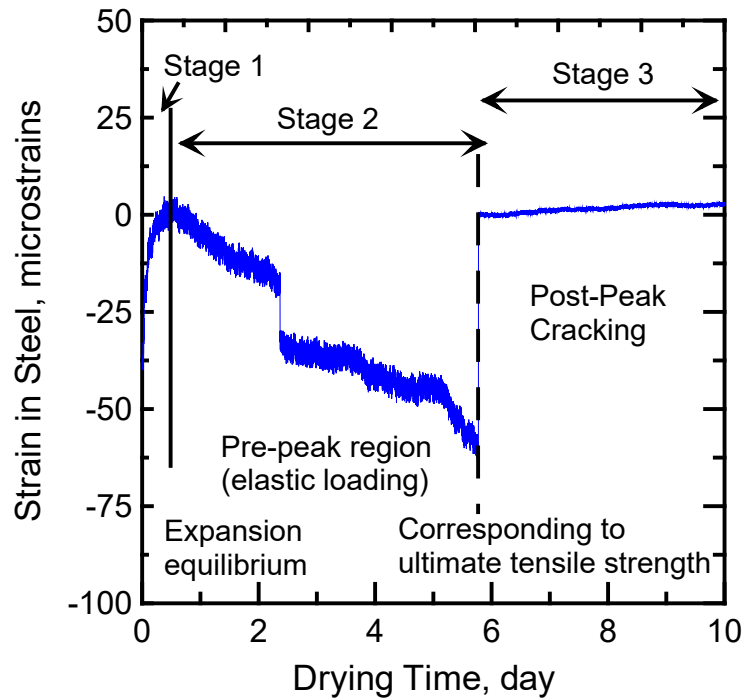
- **ACI:** Rate of Evaporation (RE) < 0.2 lb/ft²/hr (1.0 kg/m²/hr)

Multi-Scale testing and Modeling



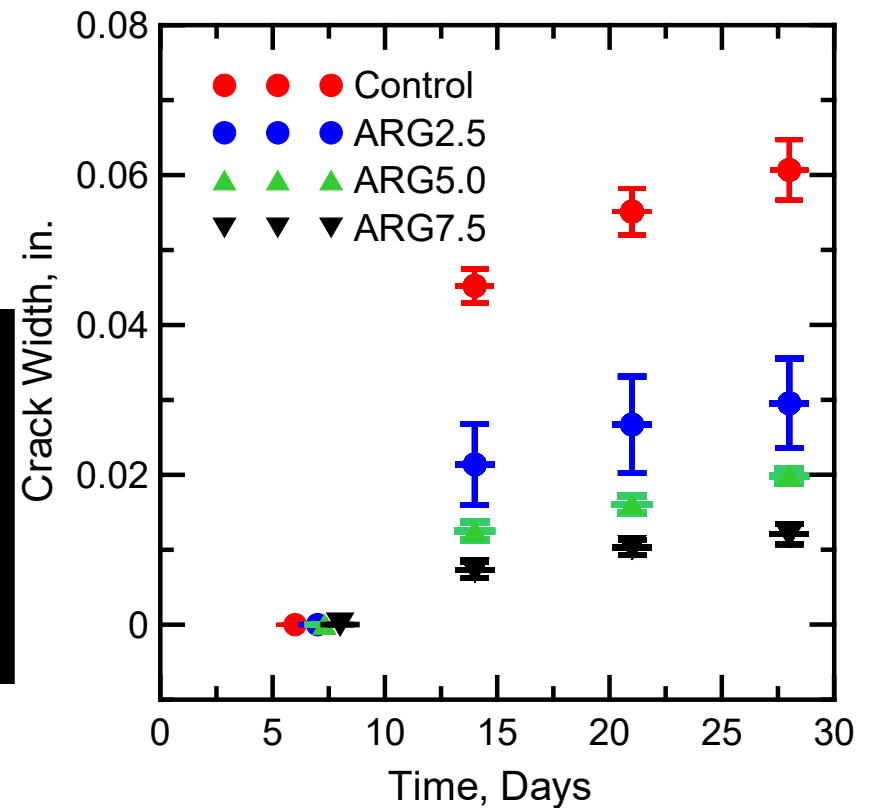
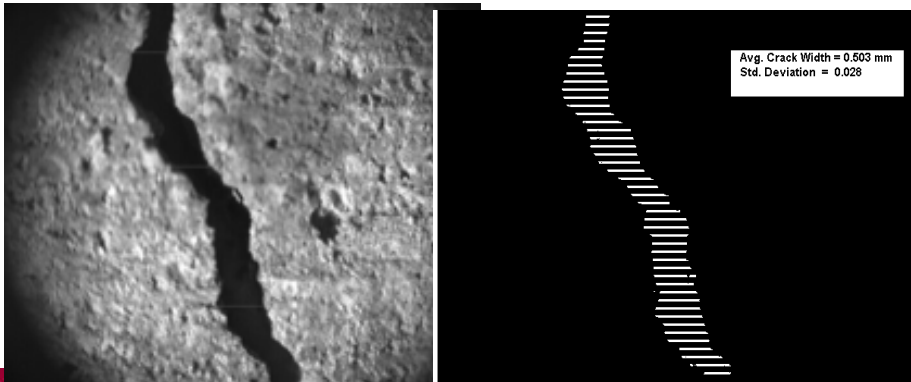
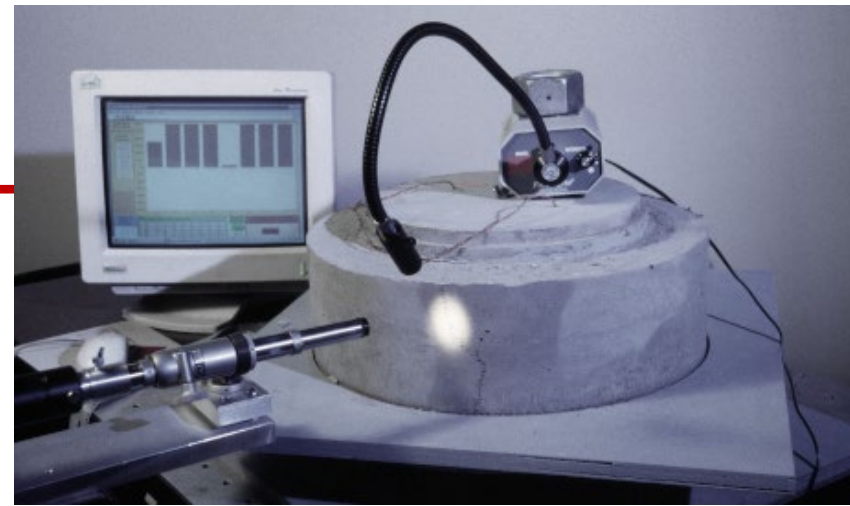
1D Shrinkage Cracking Ring Test AASHTO PP34-99

- Curing: 24 h under plastic sheets
- Relative humidity: $30\% \pm 1\%$
- Temperature: $40 \pm 1^\circ\text{C}$

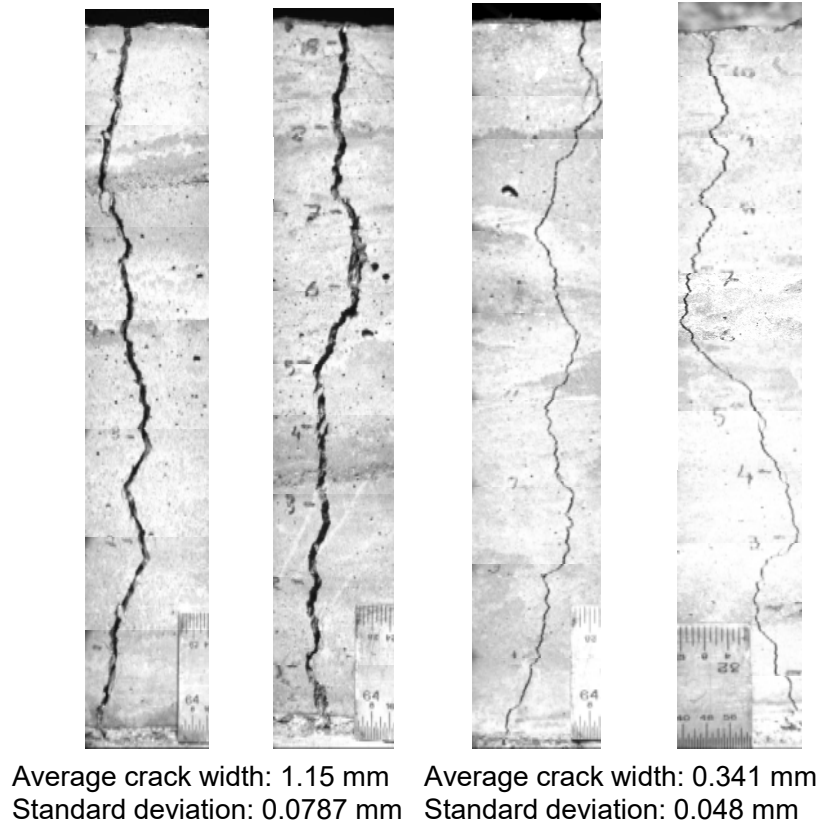
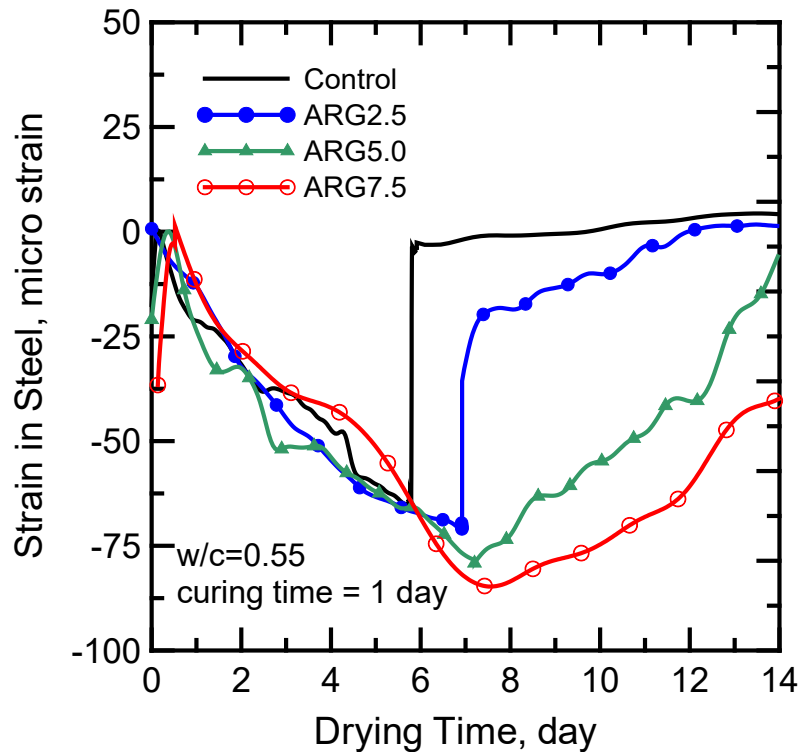


A typical result of a strain gage attached to steel ring

Image Analysis of 1D Restrained Shrinkage Cracks



Effect of fiber addition on strain in the steel rings



1. Extension of time to cracking
2. Post cracking ductility and strain distribution

Slab Curling

- Upward lift at the corner or edges
- Due to differences in moisture content or temperature
- Critical above 0.5 mm (Daiutolo 2008)
- Cause flexural cracks if a point load applied
- Additional maintenance cost

Fig. 1

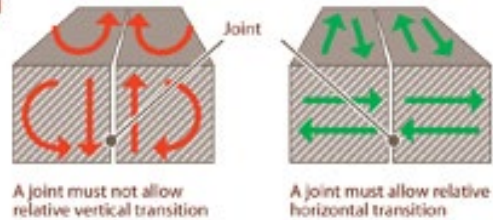


Fig. 2

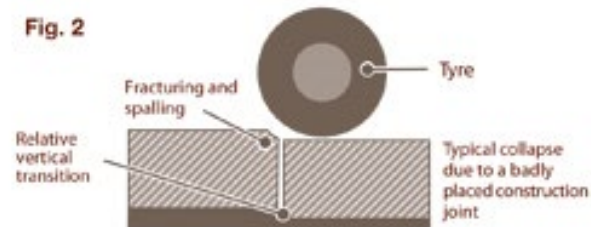
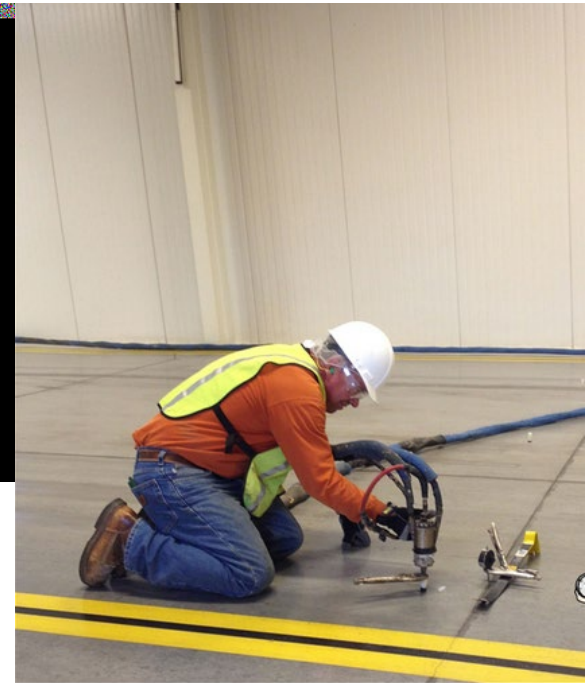
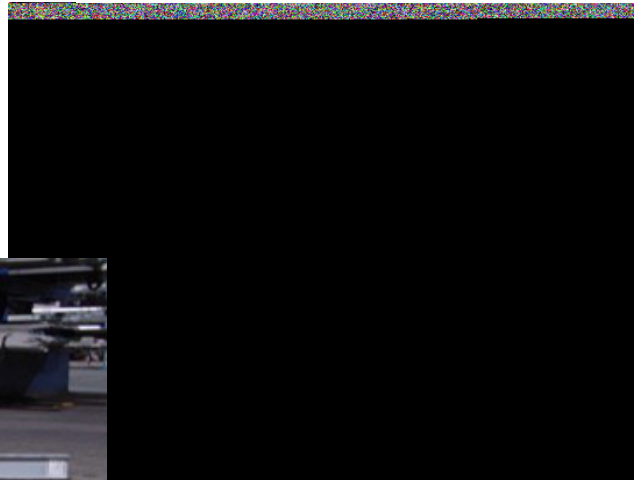


Fig. 3

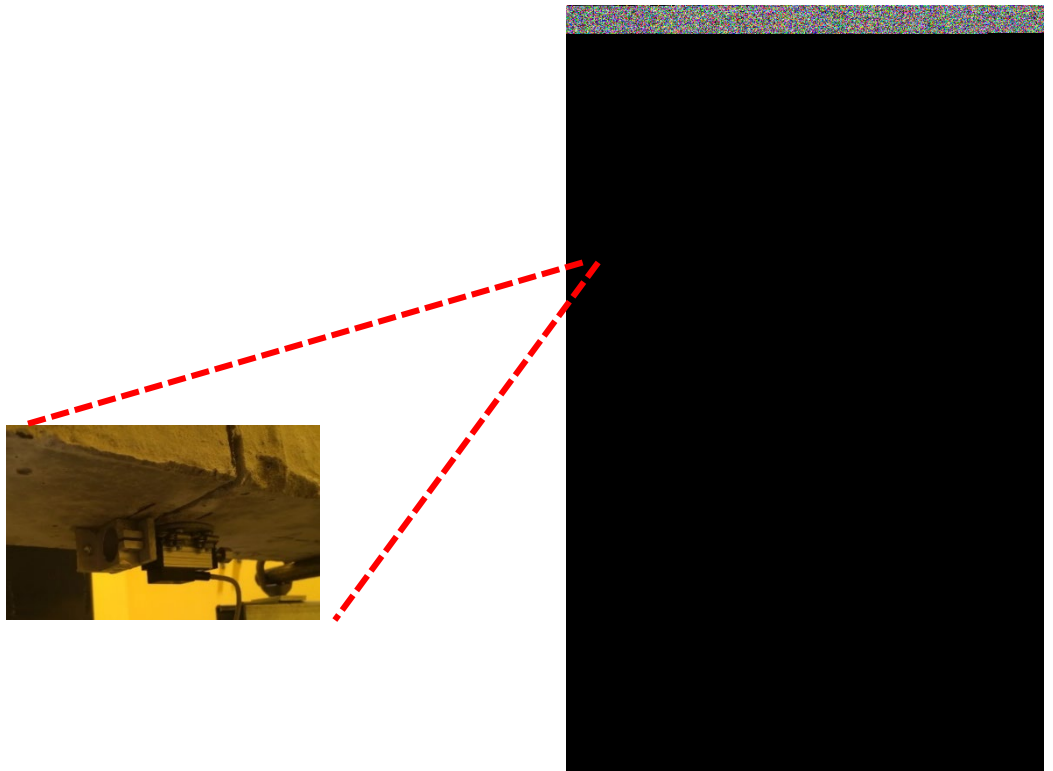


Example of construction joint damaged after a few months of life



CreepTesting Procedure

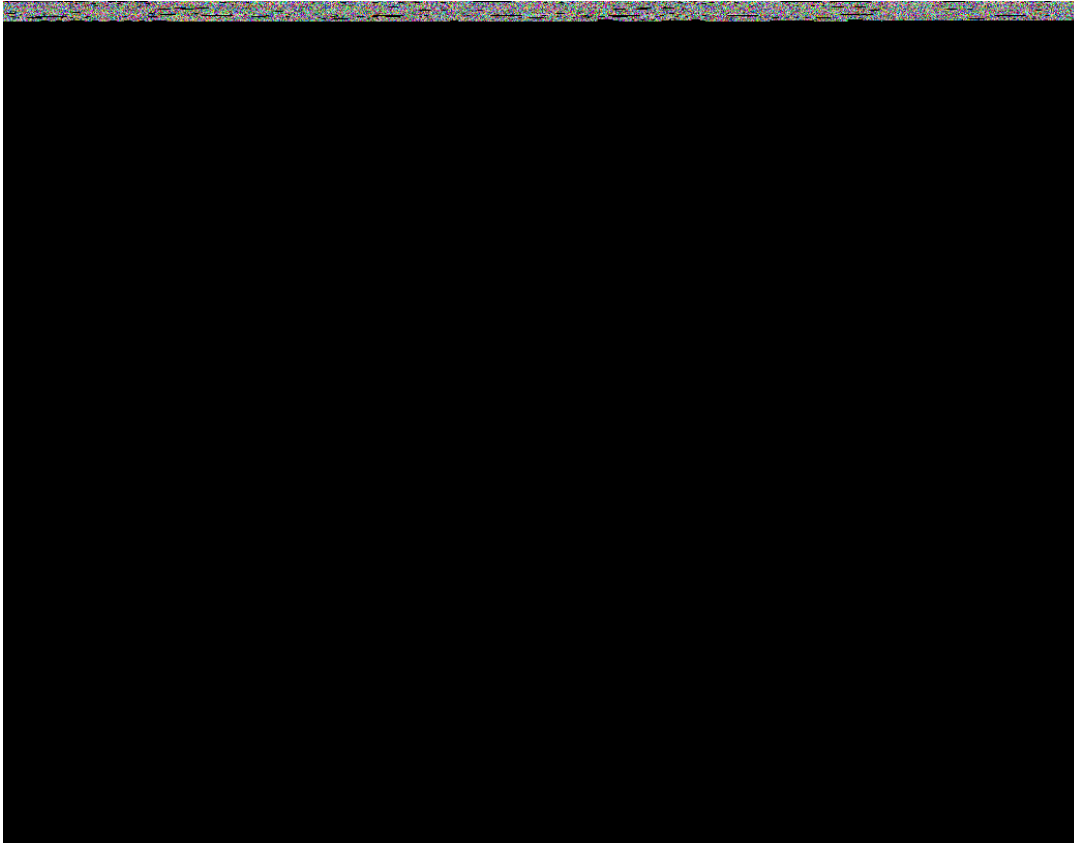
1. Fracture test up to FR_1 (CMOD = 0.5 mm/0.02 in.) using MTS machine
2. Load the cracked beam on creep testing fixture with 50% of FR_1



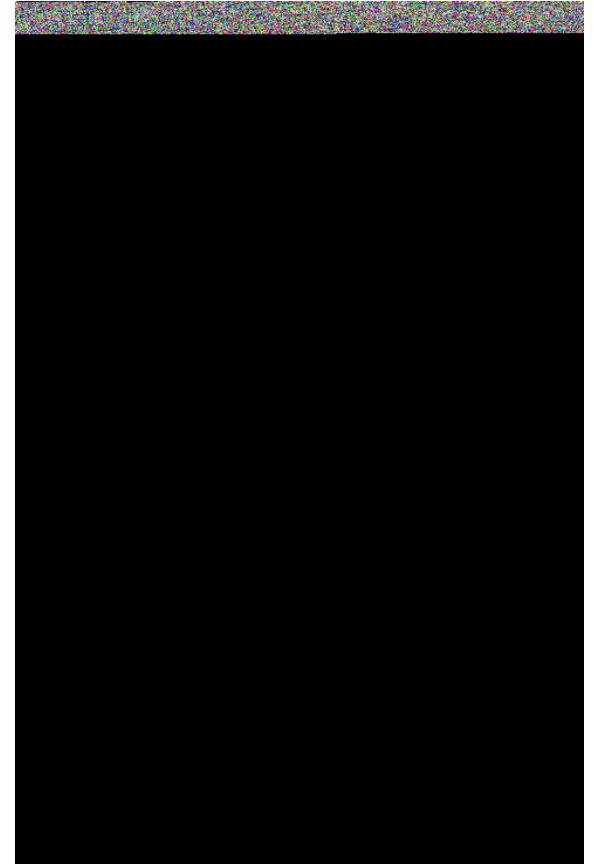
Creep Frame



Creep Test Set up

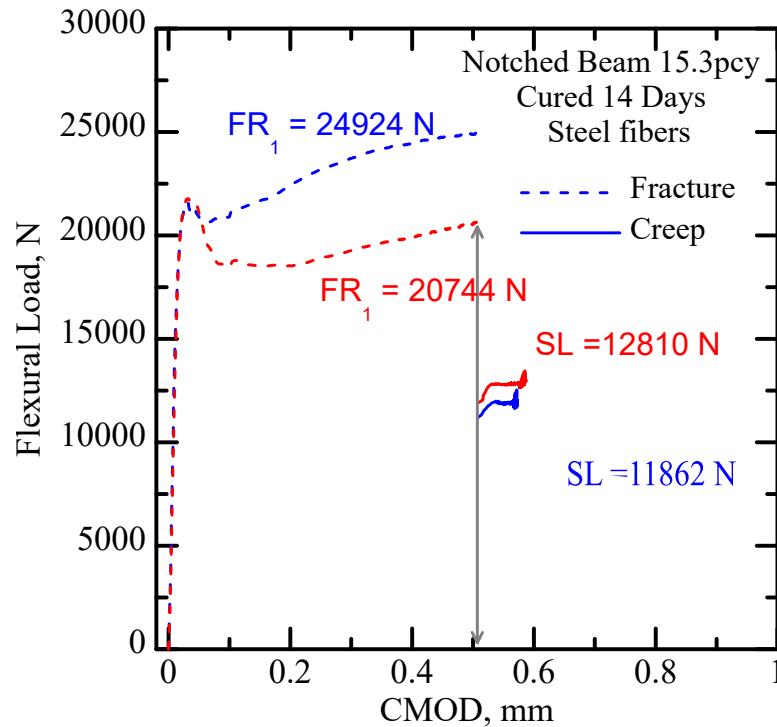


*Creep Testing Set up
Schematic Drawing*

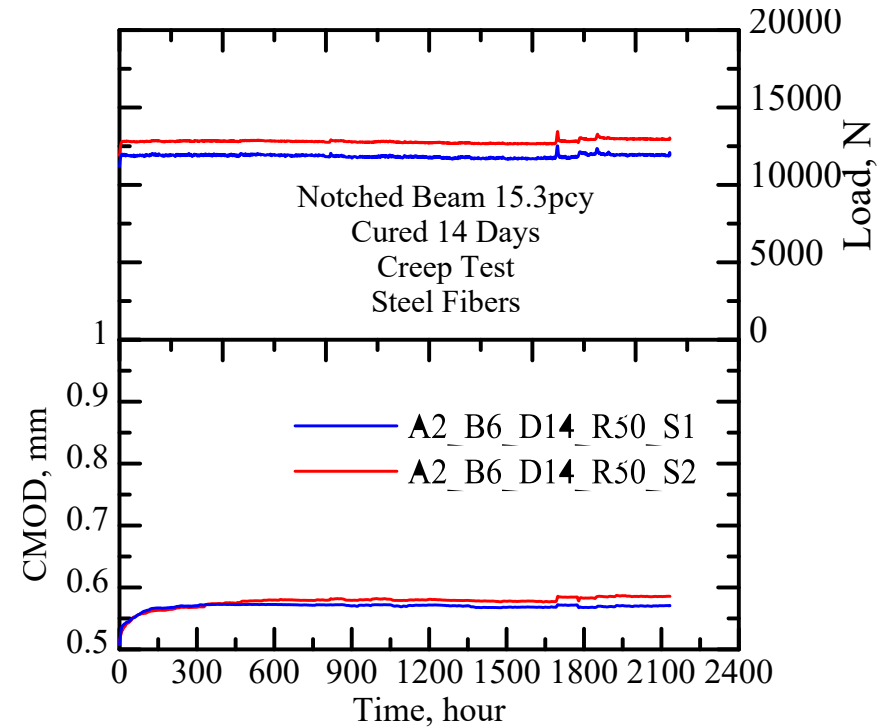


*Test Frame with
High Capacity
Piston*

Typical Creep Response



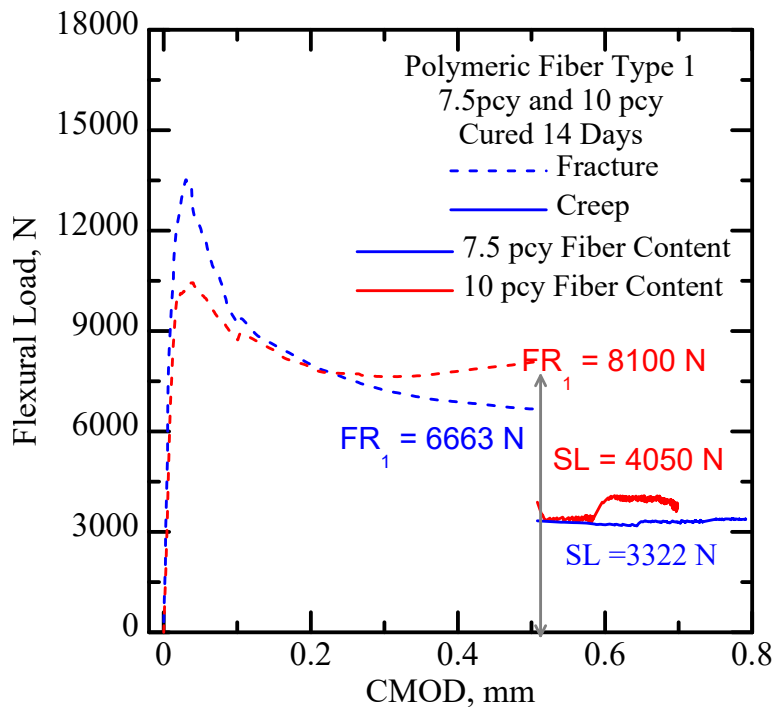
Combined Fracture and Creep versus CMOD response for a Typical Creep Test



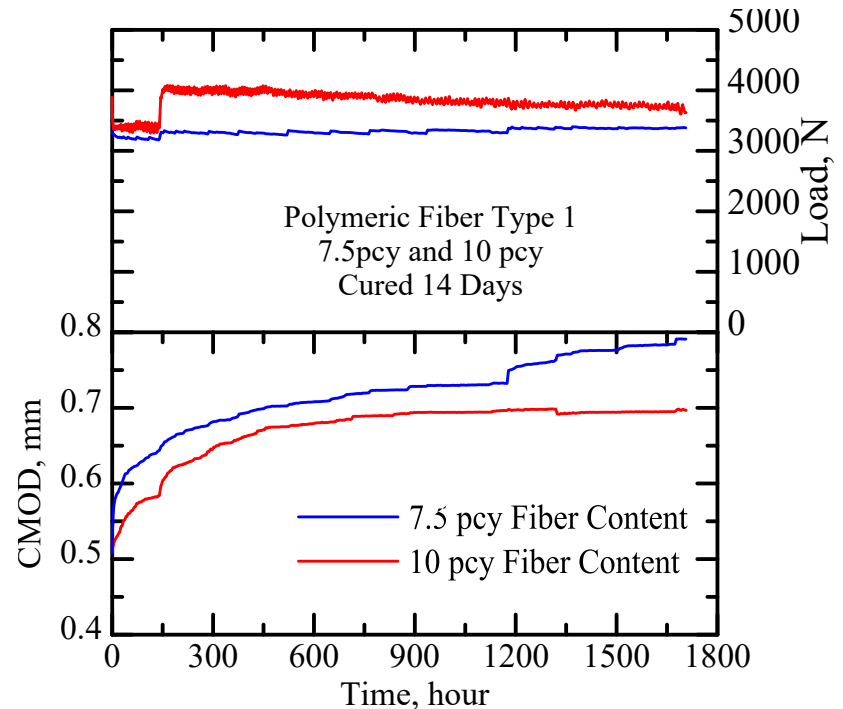
Load and CMOD versus Time response for a Typical Creep Test



Effect of Fiber Volume Fraction



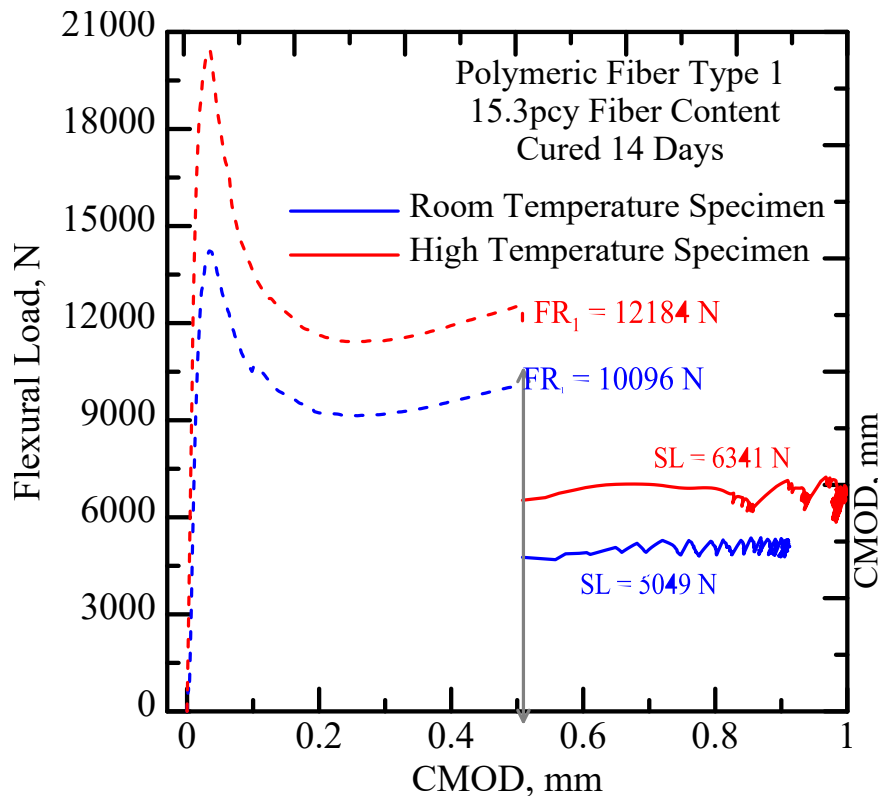
Combined Fracture and Creep versus CMOD response for a Typical Creep Test



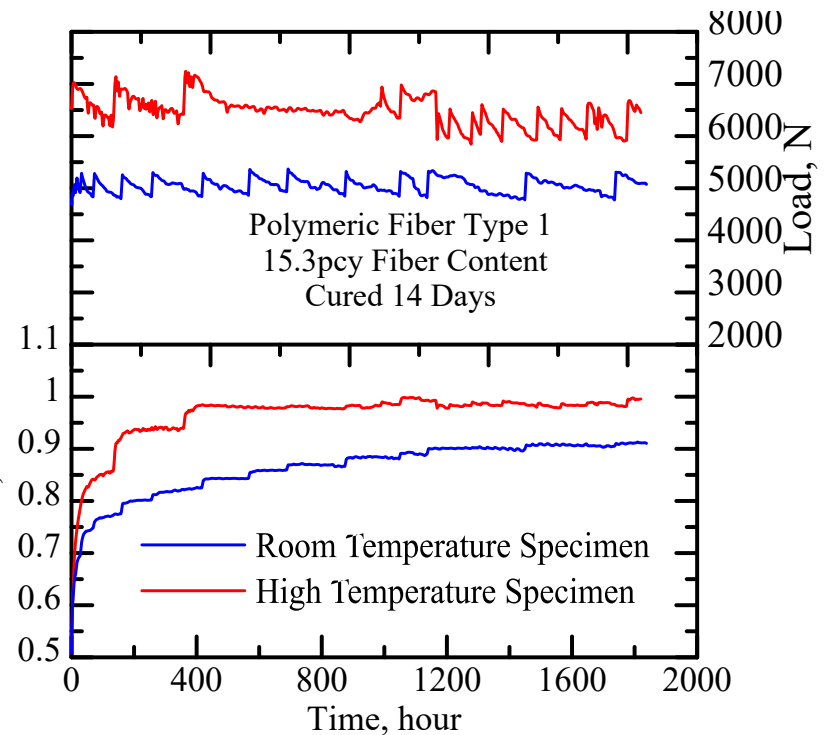
Load and CMOD versus Time response for a Typical Creep Test



Effect of Temperature



Combined Fracture and Creep versus CMOD response for a Typical Creep Test



Load and CMOD versus Time response for a Typical Creep Test